EVOLUTIONARY COMPUTATION

LABORATORY 03

Local search

for the Traveling Salesperson Problem with Node Costs

ASSIGNMENT 3 - REPORT

SOURCE REPOSITORY

https://github.com/Flat-Earther/Evolutionary-Computation/tree/main/lab-03-local-search

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OBJECTIVE

Minimize cycle length + selected node costs

Improve feasible cycles through local search neighborhoods.

The supplied instances contain 200 nodes, so every solution selects 100.

INSTANCES

TSPA and TSPB

200 nodes each - 100 selected

METHODS

8 local-search variants

200 evaluated solutions per method

Poznań University of Technology - Artificial Intelligence

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1. Problem description

The input consists of n nodes, each with integer planar coordinates (x, y) and an integer node cost.

The task is to choose exactly k = ceil(n / 2) distinct nodes and specify their cyclic visiting order.

The cycle must return from its last selected node to its first selected node.

OBJECTIVE

f(S) = cycle length(S) + sum of costs of selected nodes

DISTANCE

d(i,j) = round(sqrt((xi-xj)^2 + (yi-yj)^2))

Distances are calculated once
and stored in a distance matrix.

DATA SEPARATION

Optimization methods access
the distance matrix and node
costs, not the original coordinates.

For the supplied TSPA and TSPB instances, n = 200 and k = 100.

Local search starts from a feasible cycle and repeatedly tests a defined neighborhood. Every accepted move changes the selected-node set, the route order, or both while preserving feasibility.

Solution representation

A solution is an ordered list of k distinct node indices.

The closing edge from the last node to the first is included in the objective.

All reported indices are zero-based, matching the supplied solution checker.

2. Algorithms and pseudocode

Each variant accepts either a random feasible cycle or the best greedy construction as its starting solution S0.

D stores rounded distances, C stores node costs, and every variant also allows selected-to-unselected replacement moves.

2.1 Greedy search / node exchange

Standalone procedure for greedy improvement with intra-route swaps and selected-to-unselected replacements.

```
Input: D[i][j], C[i], k = ceil(n / 2)
Input: feasible start cycle S0
 S0 is random or best greedy; S <- S0
F(S) = sum D[S[p]][S[next(p)]] + sum C[S[p]]
swap_delta = changed edge sum after swapping i,j
replace_delta(p,v) = new two edges - old two edges
 + C[v] - C[S[p]]; v is currently unused
repeat:
 M <- all swaps (i,j) and replacements (p,v)
 shuffle M; improved <- false
 for m in M:
 delta <- swap_delta or replace_delta
 if delta < 0: apply m; F <- F + delta
 improved <- true; break
 if not improved: stop
return S, F(S)
```

2.2 Greedy search / 2-opt

Standalone procedure for greedy improvement with 2-opt route reversals and replacements.

```
Input: D[i][j], C[i], k = ceil(n / 2)
Input: feasible start cycle S0
 S0 is random or best greedy; S <- S0
F(S) = sum D[S[p]][S[next(p)]] + sum C[S[p]]
two_opt_delta(i,j) = D[a][S[j]] + D[S[i]][b]
 - D[a][S[i]] - D[S[j]][b]
 where a = previous(i), b = next(j)
replace_delta(p,v) = new two edges - old two edges
 + C[v] - C[S[p]]; v is currently unused
repeat: shuffle all 2-opt and replacement moves
 for m in shuffled moves:
 delta <- two_opt_delta or replace_delta
 if delta < 0: apply m; F <- F + delta; break
 if no move was applied: stop
return S, F(S)
```

2.3 Steepest search / node exchange

Standalone procedure for choosing the best improving swap or selected-to-unselected replacement.

```
Input: D[i][j], C[i], k = ceil(n / 2)
Input: feasible start cycle S0
 S0 is random or best greedy; S <- S0
F(S) = sum D[S[p]][S[next(p)]] + sum C[S[p]]
swap_delta = changed edge sum after swapping i,j
replace_delta(p,v) = new two edges - old two edges
 + C[v] - C[S[p]]; v is currently unused
repeat: best_delta <- 0; best_move <- none
 for every swap (i,j) and replacement (p,v):
 delta <- swap_delta or replace_delta
 if delta < best_delta: save move and delta
 if best_move is none: stop
 apply best_move; F <- F + best_delta
return S, F(S)
```

2.4 Steepest search / 2-opt

Standalone procedure for choosing the best improving 2-opt reversal or replacement.

```
Input: D[i][j], C[i], k = ceil(n / 2)
Input: feasible start cycle S0
 S0 is random or best greedy; S <- S0
F(S) = sum D[S[p]][S[next(p)]] + sum C[S[p]]
two_opt_delta(i,j) = D[a][S[j]] + D[S[i]][b]
 - D[a][S[i]] - D[S[j]][b]
 where a = previous(i), b = next(j)
replace_delta(p,v) = new two edges - old two edges
 + C[v] - C[S[p]]; v is currently unused
repeat: best_delta <- 0; best_move <- none
 for every 2-opt and replacement move:
 delta <- two_opt_delta or replace_delta
 if delta < best_delta: save move and delta
 if best_move is none: stop
 apply best_move; F <- F + best_delta
return S, F(S)
```

3. Algorithm comparison

Local-search names follow: start type / improvement rule / intra-route move.

Random start = a shuffled feasible cycle; Best greedy start = the best Lab 01/02 constructive start.

Steepest = best improving move; Greedy = first improving move. Every variant also uses selected-to-unselected exchange.

Randomization: before every greedy-search iteration, one combined neighborhood is built with all allowed intra-route moves (node exchange or 2-opt) and all selected-to-unselected replacements. The complete list is shuffled with std::shuffle using the method-specific seeded generator, so no move type is always examined first. Greedy search then accepts the first negative-delta move; after applying it, a new combined neighborhood is built and shuffled. Steepest variants use the same combined neighborhood but do not shuffle it; they evaluate every move and choose the most negative delta.

Algorithm	TSPA: average (min - max)	TSPB: average (min - max)
Random solution	260294.40(233130 - 289417)	206384.53(184618 - 235381)
Nearest neighbor - end	87476.63(84362 - 90075)	59397.34(56007 - 63166)
Nearest neighbor - any position	74346.97(72328 - 77022)	59480.28(52915 - 66164)
Greedy cycle	73793.41(72590 - 75281)	59649.91(55655 - 63734)
2-regret - path	121241.39(112968 - 131943)	73139.61(67937 - 79102)
2-regret - cycle	120502.06(111748 - 129972)	72327.22(66167 - 77274)
Weighted regret - path	75189.33(72307 - 78176)	57804.59(51301 - 65594)
Weighted regret - cycle	73335.27(72011 - 76027)	57903.04(52860 - 63406)
Random / steepest / node exchange	90222.24(82160 - 98199)	65886.03(59057 - 73129)
Random / steepest / 2-opt	77283.76(74843 - 80029)	53457.16(50828 - 56237)
Random / greedy / node exchange	87940.23(82330 - 95592)	63910.23(58625 - 71791)
Random / greedy / 2-opt	77398.37(74948 - 80421)	53797.47(51505 - 58131)
Best greedy / steepest / node exchange	**72983.04(71955 - 74861)**	53902.86(49785 - 60444)
Best greedy / steepest / 2-opt	73003.71(71955 - 74700)	**53306.10(49676 - 59789)**
Best greedy / greedy / node exchange	72984.80(71955 - 74861)	53922.85(49849 - 60444)
Best greedy / greedy / 2-opt	73015.66(71955 - 74844)	53373.11(49710 - 59864)

lowest average

Best averages in this comparison

TSPA: Best greedy / steepest / node exchange - 72983.04 (minimum 71955, maximum 74861).

TSPB: Best greedy / steepest / 2-opt - 53306.10 (minimum 49676, maximum 59789).

4. Detailed evaluation - TSPA

This page preserves the detailed statistics table used in the previous reports.

The eight Lab 03 methods use 200 random or greedy starts, depending on the variant.

Minimum, maximum, average, and standard deviation are reported for each method.

Algorithm	Minimum	Maximum	Average	Std. dev.
Random solution	233130	289417	260294.40	9730.46
Nearest neighbor - end	84362	90075	87476.63	1328.22
Nearest neighbor - any position	72328	77022	74346.97	981.62
Greedy cycle	72590	75281	73793.41	527.45
2-regret - path	112968	131943	121241.39	3928.02
2-regret - cycle	111748	129972	120502.06	3493.64
Weighted regret - path	72307	78176	75189.33	1612.85
Weighted regret - cycle	72011	76027	73335.27	869.43
Random / steepest / node exchange	82160	98199	90222.24	3156.86
Random / steepest / 2-opt	74843	80029	77283.76	1093.08
Random / greedy / node exchange	82330	95592	87940.23	2624.77
Random / greedy / 2-opt	74948	80421	77398.37	956.95
Best greedy / steepest / node exchange	**71955**	74861	**72983.04**	773.21
Best greedy / steepest / 2-opt	71955	74700	73003.71	731.49
Best greedy / greedy / node exchange	71955	74861	72984.80	769.40
Best greedy / greedy / 2-opt	71955	74844	73015.66	744.31

lowest minimum

lowest average

4. Result summary - TSPA

Result summary

Best individual result

Best greedy / steepest / node exchange with objective 71955.

Best average result

Best greedy / steepest / node exchange with average objective 72983.04.

Runtime reference

Method	Runtime
Random solution	3 ms
Nearest neighbor - end	84 ms
Nearest neighbor - any position	7174 ms
Greedy cycle	7596 ms
2-regret - path	20569 ms
2-regret - cycle	15274 ms
Weighted regret - path	20546 ms
Weighted regret - cycle	15081 ms
Random / steepest / node exchange	29987 ms
Random / steepest / 2-opt	3037 ms
Random / greedy / node exchange	9485 ms
Random / greedy / 2-opt	6600 ms
Best greedy / steepest / node exchange	857 ms
Best greedy / steepest / 2-opt	75 ms
Best greedy / greedy / node exchange	554 ms
Best greedy / greedy / 2-opt	105 ms

4. Detailed evaluation - TSPB

This page preserves the detailed statistics table used in the previous reports.

The eight Lab 03 methods use 200 random or greedy starts, depending on the variant.

Minimum, maximum, average, and standard deviation are reported for each method.

Algorithm	Minimum	Maximum	Average	Std. dev.
Random solution	184618	235381	206384.53	8859.90
Nearest neighbor - end	56007	63166	59397.34	2093.12
Nearest neighbor - any position	52915	66164	59480.28	2171.96
Greedy cycle	55655	63734	59649.91	2804.92
2-regret - path	67937	79102	73139.61	2178.64
2-regret - cycle	66167	77274	72327.22	1916.47
Weighted regret - path	51301	65594	57804.59	2791.11
Weighted regret - cycle	52860	63406	57903.04	3551.12
Random / steepest / node exchange	59057	73129	65886.03	2800.67
Random / steepest / 2-opt	50828	56237	53457.16	919.09
Random / greedy / node exchange	58625	71791	63910.23	2259.96
Random / greedy / 2-opt	51505	58131	53797.47	1049.53
Best greedy / steepest / node exchange	49785	60444	53902.86	1894.14
Best greedy / steepest / 2-opt	**49676**	59789	**53306.10**	1808.38
Best greedy / greedy / node exchange	49849	60444	53922.85	1940.36
Best greedy / greedy / 2-opt	49710	59864	53373.11	1839.02

lowest minimum

lowest average

4. Result summary - TSPB

Result summary

Best individual result

Best greedy / steepest / 2-opt with objective 49676.

Best average result

Best greedy / steepest / 2-opt with average objective 53306.10.

Runtime reference

Method	Runtime
Random solution	2 ms
Nearest neighbor - end	82 ms
Nearest neighbor - any position	7306 ms
Greedy cycle	7633 ms
2-regret - path	20435 ms
2-regret - cycle	15025 ms
Weighted regret - path	20481 ms
Weighted regret - cycle	14910 ms
Random / steepest / node exchange	29723 ms
Random / steepest / 2-opt	3542 ms
Random / greedy / node exchange	10041 ms
Random / greedy / 2-opt	7955 ms
Best greedy / steepest / node exchange	1874 ms
Best greedy / steepest / 2-opt	245 ms
Best greedy / greedy / node exchange	841 ms
Best greedy / greedy / 2-opt	346 ms

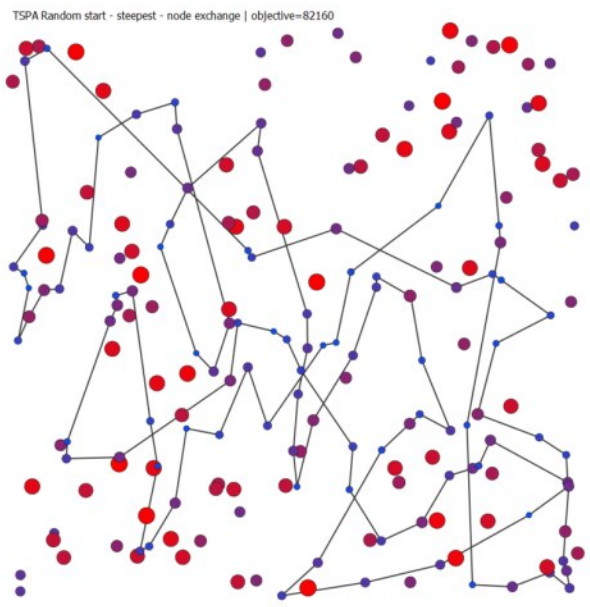
5. Best solutions - TSPA

The plots show node cost using colour and size. The line represents the closed cycle.

Random / steepest / node exchange

min=82160 max=98199 avg=90222.24

best = 82160



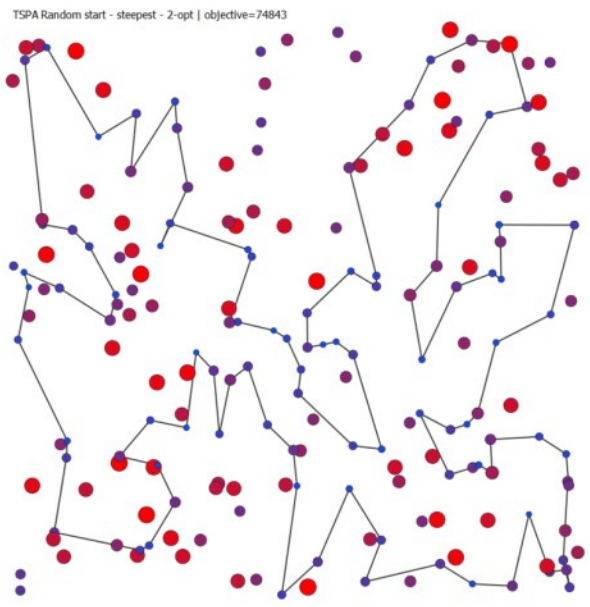
Best solution nodes:

```
107 182 192 114 181 134 176 150 174 8 9 54
131 41 135 73 140 179 52 139 112 4 76 132
180 28 156 71 157 91 80 152 108 49 11 119
168 63 149 116 60 31 48 51 16 143 85 3 50
47 93 43 130 159 146 113 89 42 151 138 158
58 84 53 144 67 35 102 161 127 97 160 83
123 90 30 187 171 2 189 141 87 185 45 177
36 33 164 178 1 169 106 57 64 88 78 193
153 75 170 107
```

Random / steepest / 2-opt

min=74843 max=80029 avg=77283.76

best = 74843



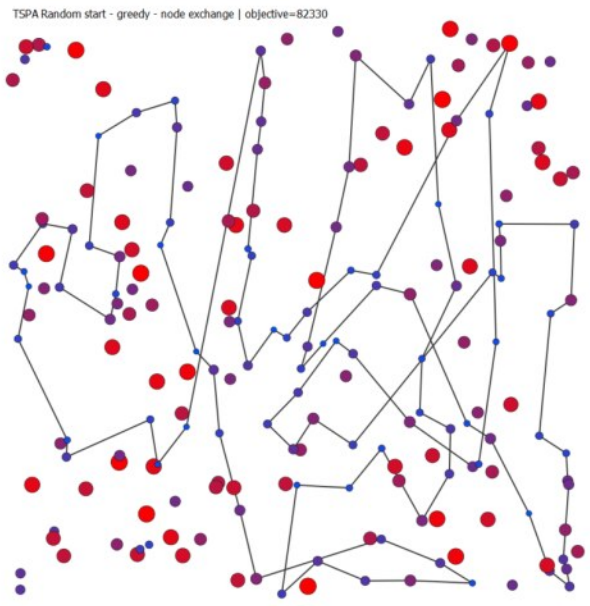
Best solution nodes:

```
135 114 181 58 158 192 182 57 64 107 88
193 78 153 146 170 159 130 113 103 89 42
151 176 150 106 43 93 47 147 1 169 50 3 85
174 8 142 164 178 101 29 163 136 186 62 9
54 33 52 179 177 36 131 134 84 140 53 144
67 35 156 16 51 76 132 4 71 157 38 91 80
48 31 60 108 152 123 160 49 149 63 168 97
127 69 171 2 189 187 161 30 141 180 28 87
102 185 45 73 135
```

Random / greedy / node exchange

min=82330 max=95592 avg=87940.23

best = 82330



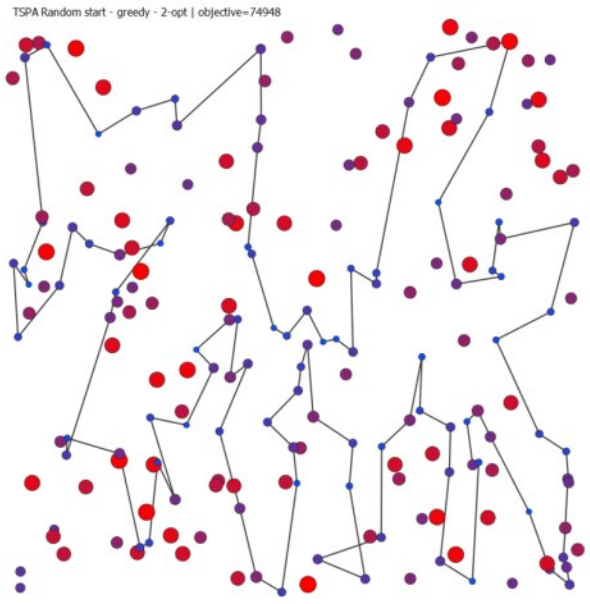
Best solution nodes:

```
89 93 178 29 184 9 33 52 144 67 185 35 16
51 139 112 5 115 141 187 30 127 97 168 63
149 116 60 108 49 160 123 100 152 80 91
157 71 76 132 180 28 87 98 25 158 182 57
111 192 181 114 135 58 134 138 151 150 176
174 85 164 136 186 154 62 143 179 53 177
54 194 8 106 113 107 88 78 193 153 170 146
159 130 47 37 147 1 169 50 3 84 41 73 45
140 36 131 72 42 89
```

Random / greedy / 2-opt

min=74948 max=80421 avg=77398.37

best = 74948



Best solution nodes:

```
78 193 153 75 146 170 159 130 93 47 147
169 1 3 50 85 164 178 29 136 186 9 54 33
131 36 177 52 144 67 16 51 139 112 115 71
157 91 80 48 31 60 149 63 116 168 49 108
152 100 132 76 123 160 127 97 161 171 2
187 189 30 141 28 180 156 35 102 185 87 98
25 114 135 73 45 140 53 179 41 84 58 192
181 158 134 72 174 176 150 151 182 57 89
42 106 43 113 107 88 78
```

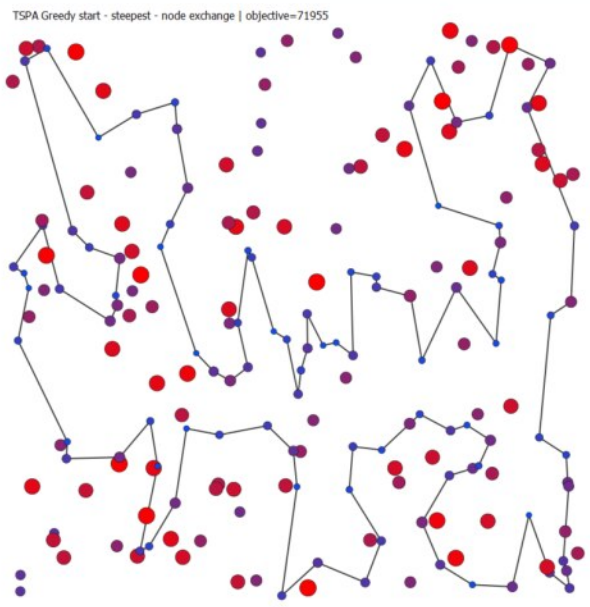
5. Best solutions - TSPA

The plots show node cost using colour and size. The line represents the closed cycle.

Best greedy / steepest / node exchan

min=71955 max=74861 avg=72983.04

best = 71955



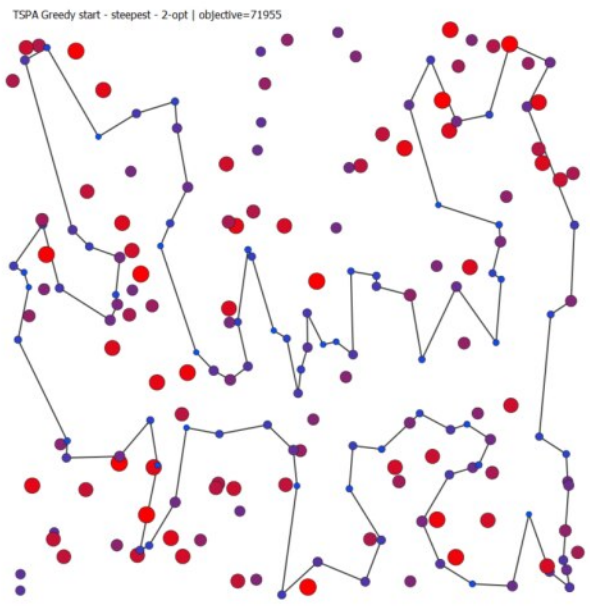
Best solution nodes:

```
101 55 29 178 184 136 186 164 1 169 3 50
93 85 174 8 54 9 33 131 36 177 52 179 53
140 144 67 16 51 35 185 102 28 180 132 76
4 71 157 91 80 48 31 108 152 100 123 83
160 49 60 116 149 63 168 97 127 161 30 187
171 2 189 141 87 45 73 135 114 181 192 158
58 84 134 72 176 150 106 113 89 42 151 138
182 57 64 107 88 78 193 153 170 146 159
130 47 37 147 101
```

Best greedy / steepest / 2-o

min=71955 max=74700 avg=73003.71

best = 71955



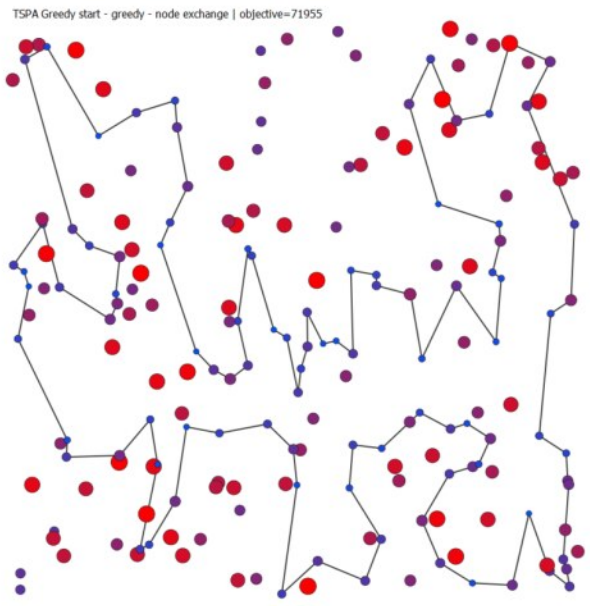
Best solution nodes:

```
55 101 147 37 47 130 159 146 170 153 193
78 88 107 64 57 182 138 151 42 89 113 106
150 176 72 134 84 58 158 192 181 114 135
73 45 87 141 189 2 171 187 30 161 127 97
168 63 149 116 60 49 160 83 123 100 152
108 31 48 80 91 157 71 4 76 132 180 28 102
185 35 51 16 67 144 140 53 179 52 177 36
131 33 9 54 8 174 85 93 50 3 169 1 164 186
136 184 178 29 55
```

Best greedy / greedy / node exchang

min=71955 max=74861 avg=72984.80

best = 71955



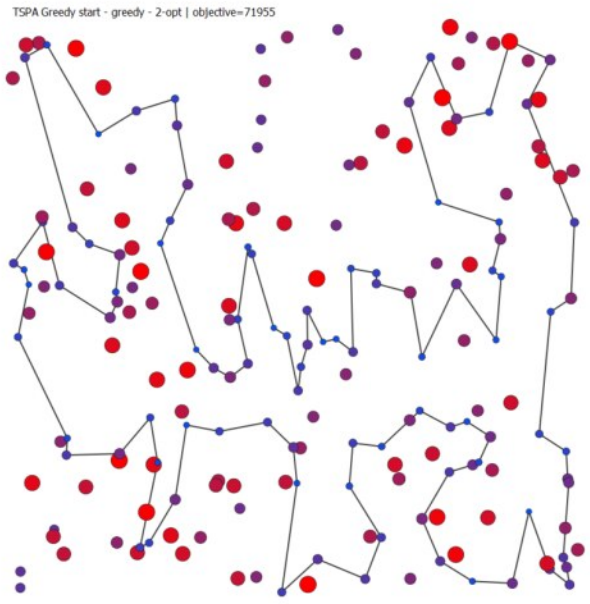
Best solution nodes:

```
101 55 29 178 184 136 186 164 1 169 3 50
93 85 174 8 54 9 33 131 36 177 52 179 53
140 144 67 16 51 35 185 102 28 180 132 76
4 71 157 91 80 48 31 108 152 100 123 83
160 49 60 116 149 63 168 97 127 161 30 187
171 2 189 141 87 45 73 135 114 181 192 158
58 84 134 72 176 150 106 113 89 42 151 138
182 57 64 107 88 78 193 153 170 146 159
130 47 37 147 101
```

Best greedy / greedy / 2-opt

min=71955 max=74844 avg=73015.66

best = 71955



Best solution nodes:

```
55 101 147 37 47 130 159 146 170 153 193
78 88 107 64 57 182 138 151 42 89 113 106
150 176 72 134 84 58 158 192 181 114 135
73 45 87 141 189 2 171 187 30 161 127 97
168 63 149 116 60 49 160 83 123 100 152
108 31 48 80 91 157 71 4 76 132 180 28 102
185 35 51 16 67 144 140 53 179 52 177 36
131 33 9 54 8 174 85 93 50 3 169 1 164 186
136 184 178 29 55
```

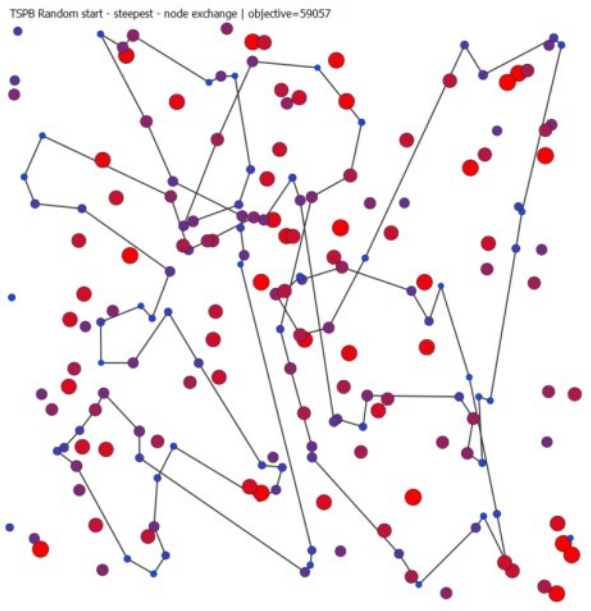
5. Best solutions - TSPB

The plots show node cost using colour and size. The line represents the closed cycle.

Random / steepest / node exchange

min=59057 max=73129 avg=65886.03

best = 59057



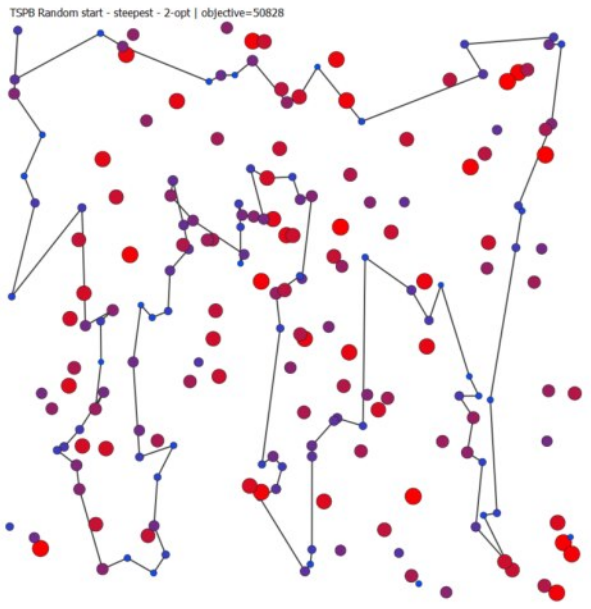
Best solution nodes:

```
180 6 29 132 197 175 134 45 157 60 40 85
51 167 59 64 47 97 170 125 154 140 38 158
145 28 149 191 21 189 8 117 72 46 34 24 67
66 80 52 182 153 129 90 10 114 105 94 70
190 76 27 111 122 173 184 3 135 162 192 54
127 14 2 20 74 171 99 181 152 98 151 86 79
57 89 48 63 30 183 133 120 96 198 121 42
147 116 50 130 136 172 160 69 179 87 82 71
155 115 180
```

Random / steepest / 2-opt

min=50828 max=56237 avg=53457.16

best = 50828



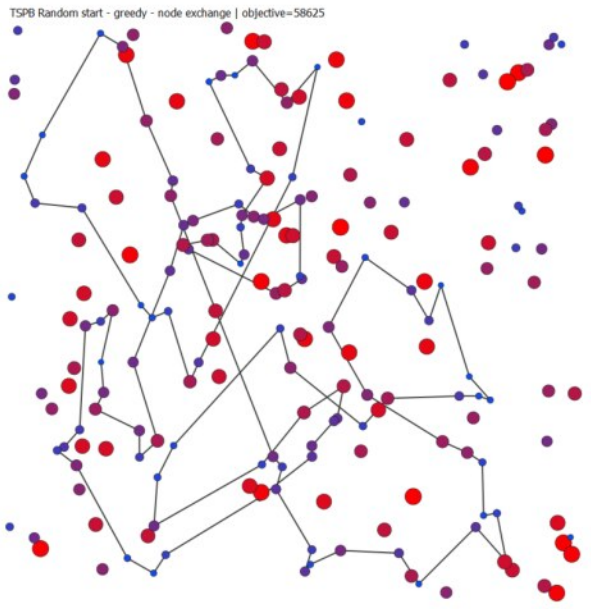
Best solution nodes:

```
196 61 105 81 94 190 70 57 89 32 156 48 96
198 121 147 116 146 37 13 149 28 145 65
158 177 26 125 170 180 115 6 29 132 197 36
62 175 134 45 157 60 40 71 97 154 140 47
38 21 30 130 191 183 160 69 172 133 136 8
117 120 92 72 46 86 3 184 122 151 59 73
167 51 85 82 87 179 181 152 34 24 67 76
162 192 54 127 129 80 52 182 153 74 2 20
14 90 10 114 196
```

Random / greedy / node exchange

min=58625 max=71791 avg=63910.23

best = 58625



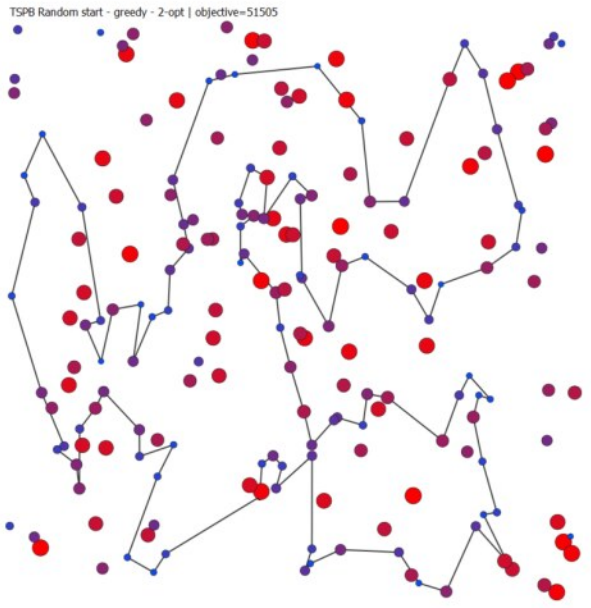
Best solution nodes:

```
155 71 141 97 38 30 183 133 8 117 46 184 3
173 122 21 176 189 168 69 160 172 136 92
120 121 198 96 48 156 89 72 64 33 47 140
154 158 145 28 149 116 147 50 130 191 73
167 51 179 82 87 88 99 171 74 2 14 20 153
182 150 66 27 76 162 192 54 127 90 129 80
84 67 98 151 40 60 157 59 25 11 24 34 152
181 85 45 134 175 197 132 29 6 177 125 26
170 115 180 155
```

Random / greedy / 2-opt

min=51505 max=58131 avg=53797.47

best = 51505



Best solution nodes:

```
34 24 67 66 84 150 80 127 90 129 52 153 14
20 2 74 17 171 99 88 87 82 179 181 51 167
73 59 85 45 134 175 60 40 71 155 115 180 6
36 197 132 29 178 65 145 28 149 158 125
177 170 26 154 97 140 47 38 21 30 130 121
96 89 57 185 104 194 190 70 187 196 114 10
78 54 192 162 76 135 27 184 3 46 86 72 117
92 120 133 136 8 172 69 160 122 151 98 25
152 34
```

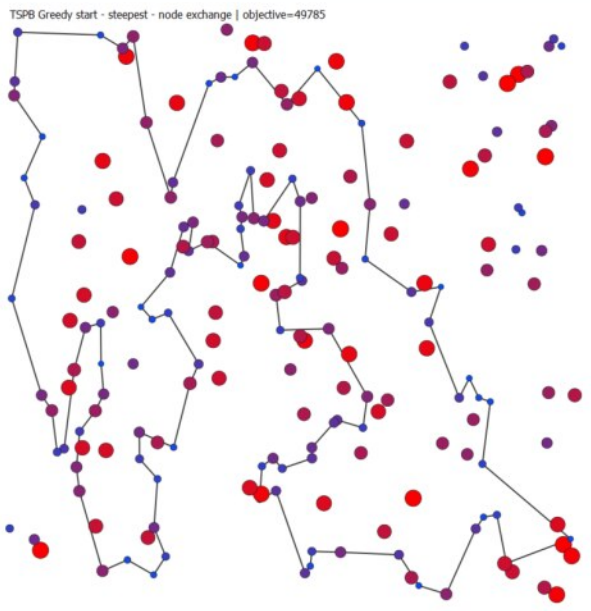
5. Best solutions - TSPB

The plots show node cost using colour and size. The line represents the closed cycle.

Best greedy / steepest / node exchan

min=49785 max=60444 avg=53902.86

best = 49785



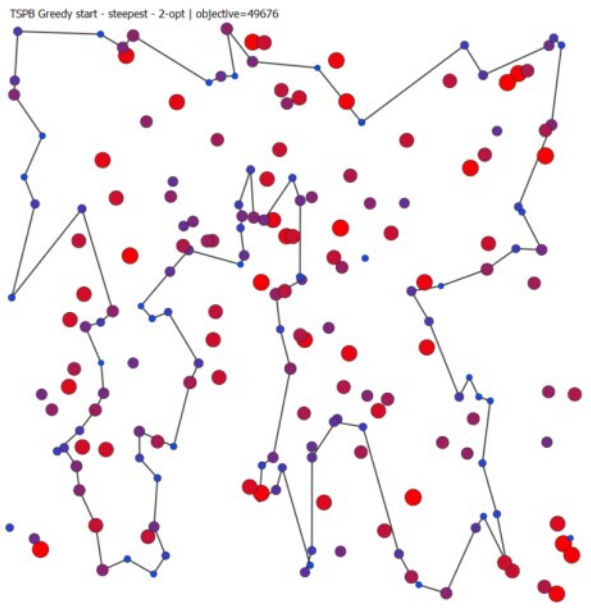
Best solution nodes:

```
110 2 14 20 74 17 171 99 88 179 87 82 51
85 59 73 167 181 152 34 24 67 66 27 151
122 3 184 46 72 117 8 120 133 136 172 160
69 189 21 183 30 38 154 140 47 64 33 40
155 71 60 157 45 134 175 62 36 197 6 180
115 170 125 177 19 29 132 103 178 65 145
28 149 13 146 37 116 147 42 50 191 130 121
198 96 48 156 89 57 185 76 162 54 192 80
127 129 90 153 110
```

Best greedy / steepest / 2-o

min=49676 max=59789 avg=53306.10

best = 49676



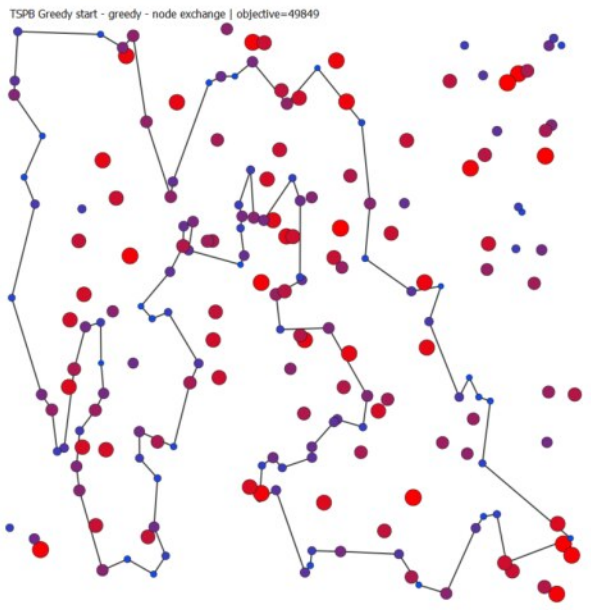
Best solution nodes:

```
61 196 114 148 10 78 54 162 192 80 127 129
90 153 14 2 20 74 17 171 39 99 67 34 24
181 179 82 87 167 51 85 59 73 98 151 122 3
184 46 72 117 8 120 133 136 172 160 69 21
38 154 140 47 64 33 40 155 71 60 157 45
134 175 62 91 36 197 29 132 6 180 115 170
177 125 26 158 65 145 28 149 13 146 37 116
147 42 121 198 96 58 48 89 57 190 70 81 94
105 61
```

Best greedy / greedy / node exchang

min=49849 max=60444 avg=53922.85

best = 49849



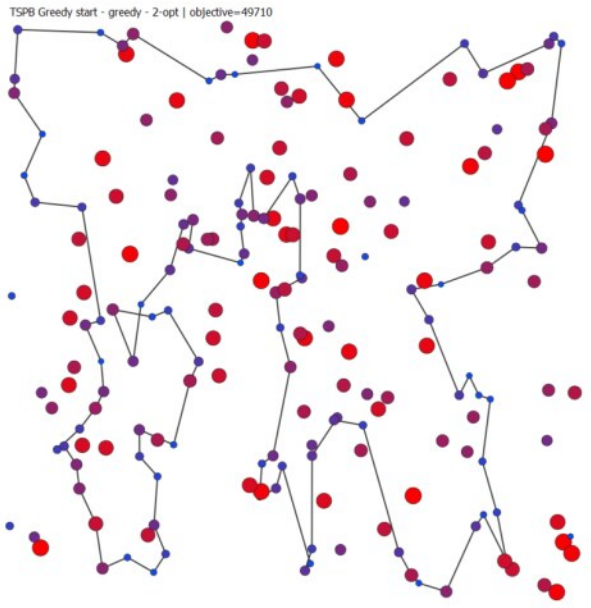
Best solution nodes:

```
110 2 14 20 74 17 171 99 88 179 87 82 51
85 59 73 167 181 152 34 24 67 66 27 151
122 3 184 46 72 117 8 120 133 136 172 160
69 21 183 30 176 38 154 140 47 64 33 40
155 71 60 157 45 134 175 62 36 197 6 180
115 170 125 177 19 29 132 103 178 65 145
28 149 13 146 37 116 147 42 50 191 130 121
198 96 48 156 89 57 185 76 162 54 192 80
127 129 90 153 110
```

Best greedy / greedy / 2-opt

min=49710 max=59864 avg=53373.11

best = 49710



Best solution nodes:

```
38 154 97 26 140 47 64 33 40 155 71 60 157
45 134 175 62 91 36 197 29 132 6 180 115
170 177 125 158 145 28 149 13 146 37 116
147 42 121 198 96 89 57 190 70 81 94 105
61 196 114 148 10 78 54 162 192 80 127 129
90 153 14 2 20 74 17 171 39 99 67 34 24
181 179 82 87 167 51 85 59 73 98 151 122 3
184 46 72 117 8 120 133 136 172 160 69 21
183 30 38
```

6. Conclusions

Local search improves the constructive starting solutions by repeatedly accepting objective-decreasing neighborhood moves.
The observed result depends on the starting-solution source, search policy, and intra-route neighborhood.

Main conclusions

The best local-search average for TSPA was obtained by Best greedy / steepest / node exchange (72983.04), while TSPB was best with Best greedy / steepest / 2-opt (53306.10).

Steepest search evaluates the whole neighborhood before moving, whereas greedy search accepts the first improving move in a randomized order. This creates a direct trade-off between search intensity and runtime.

The 2-opt neighborhood changes route order while preserving the selected set. Node exchange adds a different intra-route ordering neighborhood, and inter-route exchange can replace a selected node with an unused node and therefore changes the selected subset.

Greedy starts based on the best previous constructive methods are consistently stronger than random starts. Local search nevertheless improves both start types, showing that the neighborhood design contributes beyond construction quality.

All generated routes passed structural validation and the delta-accounting assertion. The complete comparison table makes the improvement over Labs 01 and 02 directly visible.

SOURCE REPOSITORY

https://github.com/Flat-Earther/Evolutionary-Computation/tree/main/lab-03-local-search

The large-format best and worst local-search visualizations are presented in the appendix.

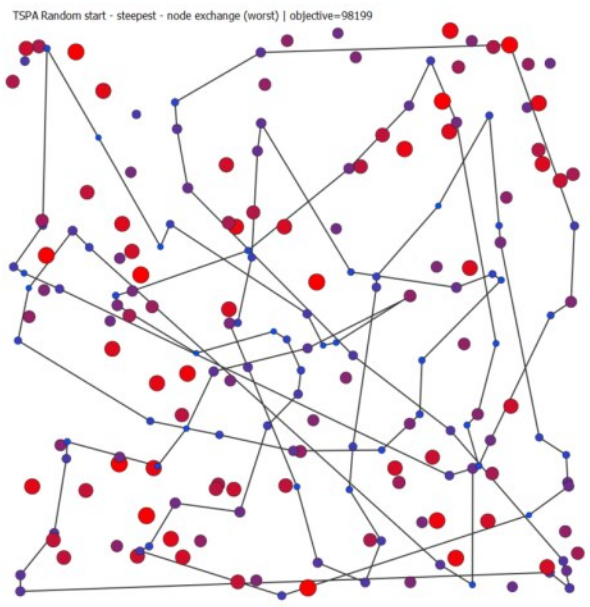
7. Worst solutions - TSPA

The plots show node cost using colour and size. The line represents the closed cycle.

Random / steepest / node exchange

min=82160 max=98199 avg=90222.24

worst = 98199



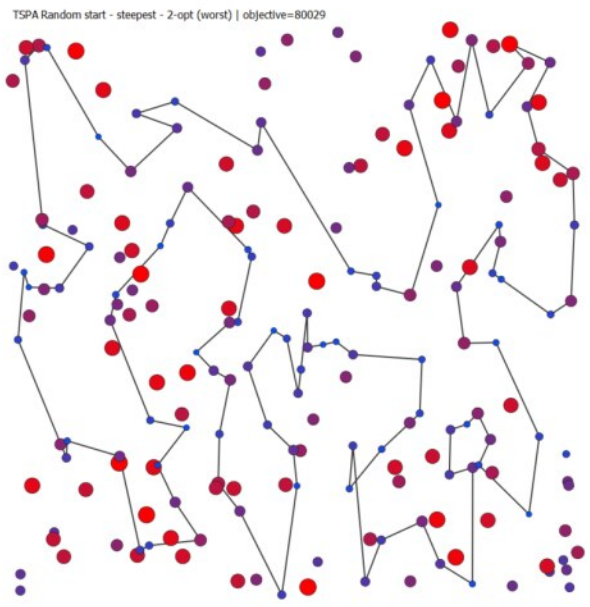
Worst solution nodes:

```
194 8 179 28 141 187 97 127 69 129 104 88
78 153 150 131 4 71 157 115 29 147 37 47
113 89 106 43 93 184 136 186 62 51 90 123
83 180 67 144 53 140 45 98 189 2 171 114
107 170 146 159 130 169 1 178 164 9 54 84
58 158 192 181 135 185 156 35 16 139 112
33 85 3 50 174 176 134 73 87 30 168 63 108
152 182 57 42 151 49 149 116 60 48 80 132
76 52 177 36 194
```

Random / steepest / 2-opt

min=74843 max=80029 avg=77283.76

worst = 80029



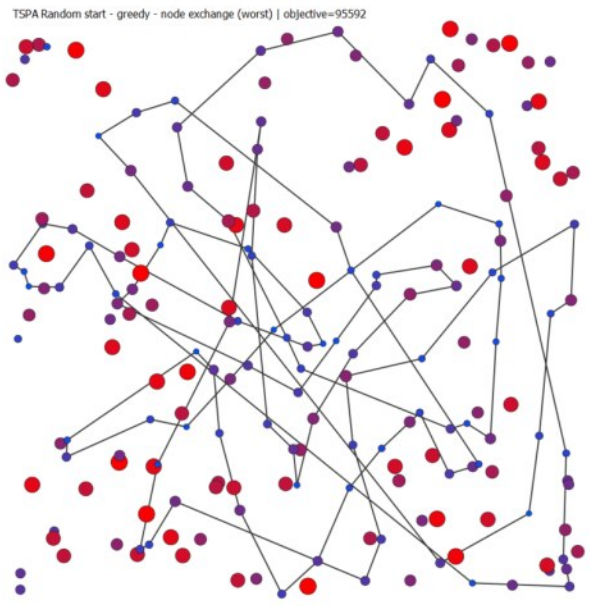
Worst solution nodes:

```
8 164 186 136 184 163 178 17 29 55 101 46
21 147 37 47 50 3 169 1 85 81 93 130 107
89 113 43 106 150 151 42 57 182 138 158
192 84 58 134 176 174 131 36 177 179 52 53
140 144 67 185 45 73 135 114 98 27 87 102
28 180 156 35 16 51 4 76 132 123 160 30
141 187 189 10 2 171 161 97 127 168 149 63
49 152 60 31 7 48 80 38 71 91 157 139 112
33 9 54 8
```

Random / greedy / node exchange

min=82330 max=95592 avg=87940.23

worst = 95592



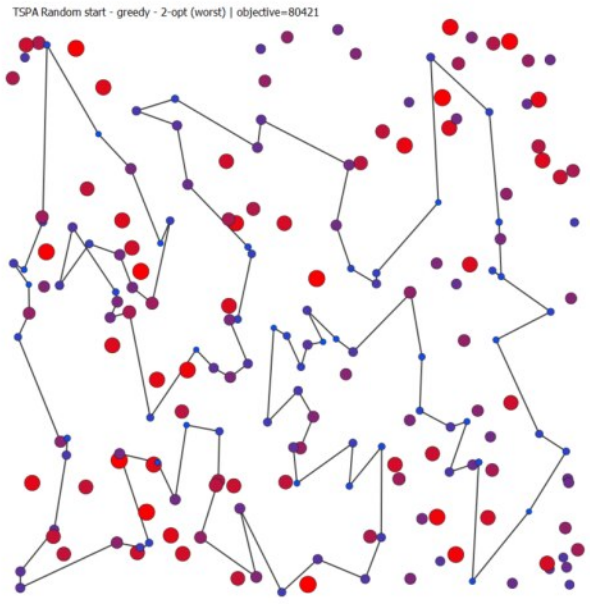
Worst solution nodes:

```
185 140 36 54 9 142 85 8 194 131 41 135 73
45 16 139 112 156 187 171 2 189 192 158 84
99 174 3 147 37 47 130 107 182 38 80 91
157 143 33 89 42 151 176 72 134 58 181 114
98 87 28 180 97 127 30 141 102 67 164 1
169 50 93 113 106 150 53 144 196 4 71 115
39 186 136 178 159 153 193 78 57 123 152
49 63 149 116 60 108 100 35 179 177 52 51
76 132 90 83 185
```

Random / greedy / 2-opt

min=74948 max=80421 avg=77398.37

worst = 80421



Worst solution nodes:

```
168 119 63 116 149 60 48 80 38 132 76 162
90 100 152 49 108 123 83 160 148 30 180 28
102 185 156 35 16 51 4 71 91 157 139 112
62 143 33 54 9 164 136 178 1 50 3 47 93
130 159 107 57 89 151 106 150 176 174 8
131 36 52 177 179 53 144 67 45 140 41 61
73 135 84 58 134 158 192 181 114 98 25 10
27 87 141 189 187 161 2 32 171 92 104 129
69 127 97 95 168
```

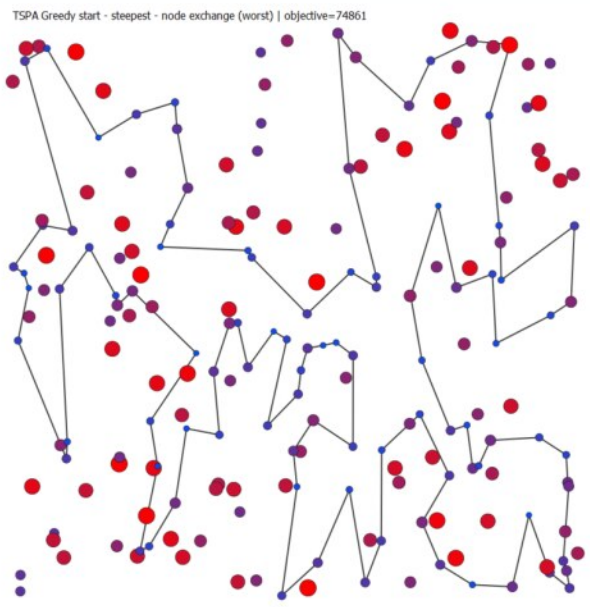
7. Worst solutions - TSPA

The plots show node cost using colour and size. The line represents the closed cycle.

Best greedy / steepest / node exchan

min=71955 max=74861 avg=72983.04

worst = 74861



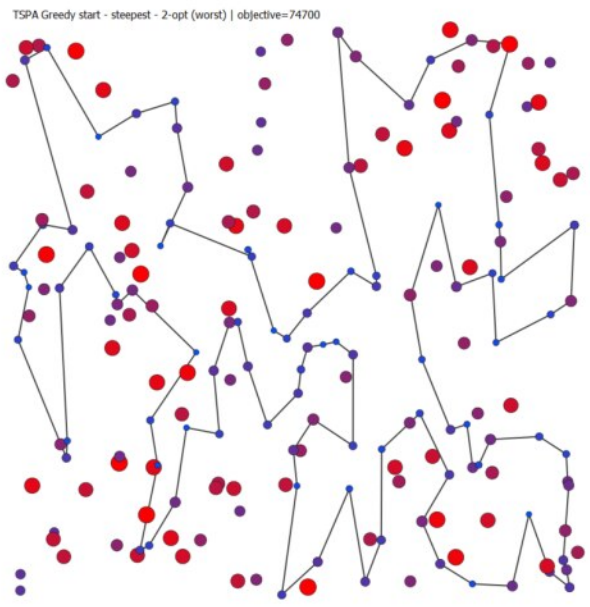
Worst solution nodes:

```
154 186 136 163 29 178 1 169 50 147 37 47
93 3 85 164 8 174 150 106 42 89 113 130
159 146 170 153 193 78 88 107 64 57 182
138 151 176 72 134 158 192 58 181 114 135
73 41 84 131 36 177 179 53 140 45 144 67
185 35 156 28 87 141 189 2 171 187 30 180
90 83 123 152 49 97 127 168 63 149 116 60
108 31 48 80 91 157 71 4 76 132 51 16 52
33 54 9 62 39 154
```

Best greedy / steepest / 2-o

min=71955 max=74700 avg=73003.71

worst = 74700



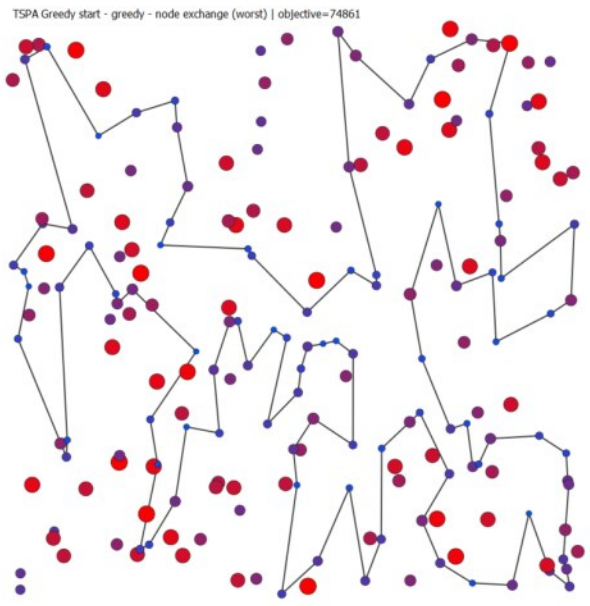
Worst solution nodes:

```
39 62 9 54 33 52 144 67 51 16 76 132 4 71
157 91 80 48 31 108 60 116 149 63 168 127
97 49 152 123 83 90 180 30 187 171 2 189
141 87 28 156 35 185 45 140 53 179 177 36
131 84 41 73 135 114 181 58 192 158 134 72
176 151 138 182 57 64 107 88 78 193 153
170 146 159 130 113 89 42 106 150 174 8
164 85 3 93 47 37 147 50 169 1 178 29 163
136 186 154 39
```

Best greedy / greedy / node exchang

min=71955 max=74861 avg=72984.80

worst = 74861



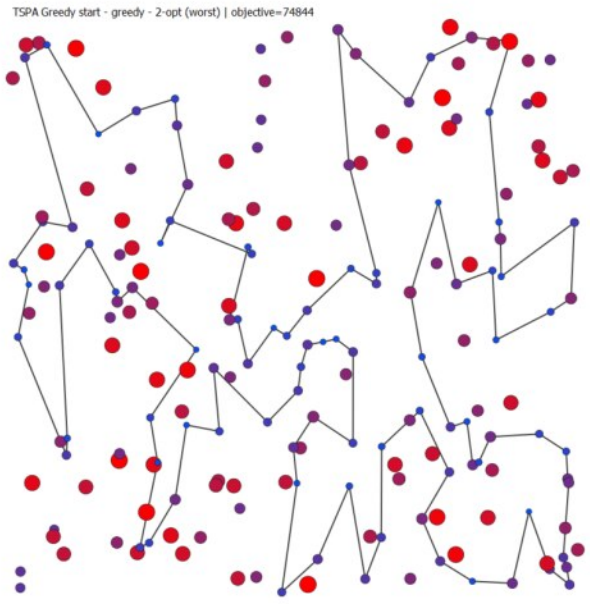
Worst solution nodes:

```
154 186 136 163 29 178 1 169 50 147 37 47
93 3 85 164 8 174 150 106 42 89 113 130
159 146 170 153 193 78 88 107 64 57 182
138 151 176 72 134 158 192 58 181 114 135
73 41 84 131 36 177 179 53 140 45 144 67
185 35 156 28 87 141 189 2 171 187 30 180
90 83 123 152 49 97 127 168 63 149 116 60
108 31 48 80 91 157 71 4 76 132 51 16 52
33 54 9 62 39 154
```

Best greedy / greedy / 2-opt

min=71955 max=74844 avg=73015.66

worst = 74844



Worst solution nodes:

```
39 62 9 54 33 52 144 67 185 35 156 51 16
76 132 4 71 157 91 80 48 31 108 60 116 149
63 168 127 97 49 152 123 83 90 180 30 187
171 2 189 141 87 28 45 140 53 179 177 36
131 84 41 73 135 114 181 58 192 158 134 72
176 151 138 182 57 64 107 88 78 193 153
170 146 159 130 113 89 42 106 150 174 8
164 85 3 93 47 37 147 50 169 1 178 29 163
136 186 154 39
```

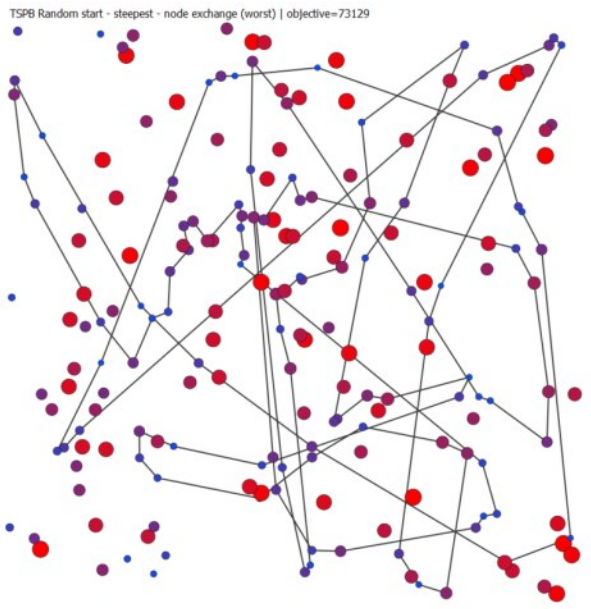
7. Worst solutions - TSPB

The plots show node cost using colour and size. The line represents the closed cycle.

Random / steepest / node exchange

min=59057 max=73129 avg=65886.03

worst = 73129



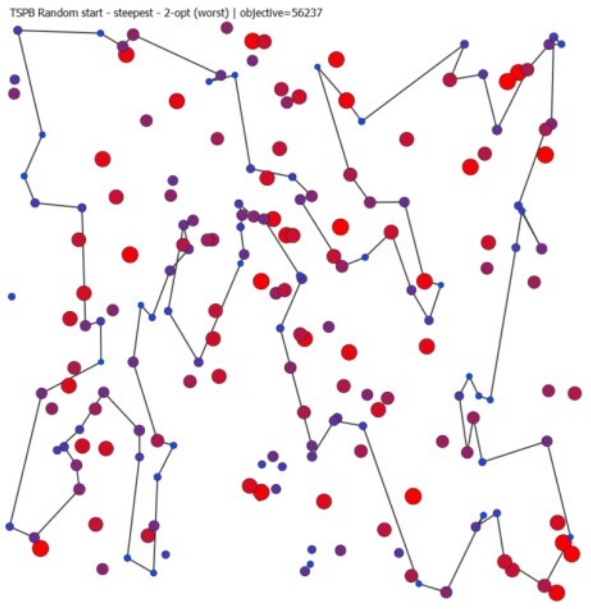
Worst solution nodes:

```
20 14 153 69 160 172 136 133 189 183 30 21
38 47 140 97 125 145 28 13 146 149 154 64
152 2 110 148 114 196 187 89 96 198 121
170 132 29 6 70 81 94 105 54 192 99 171 17
182 150 67 181 85 60 71 155 40 59 80 127
84 66 24 34 76 104 194 190 57 185 135 3
184 173 122 151 98 87 82 167 120 48 156
162 129 90 118 56 143 10 86 46 72 117 8 73
51 179 88 74 20
```

Random / steepest / 2-opt

min=50828 max=56237 avg=53457.16

worst = 56237



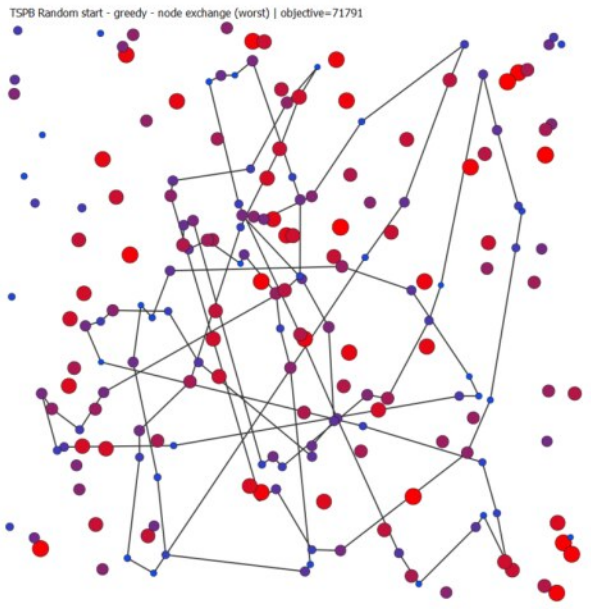
Worst solution nodes:

```
139 196 148 114 10 90 129 127 80 182 52
153 118 110 93 2 188 14 74 20 17 171 67 34
181 152 151 3 184 117 8 133 136 172 160 69
64 47 30 21 38 140 154 97 141 40 60 134
175 71 155 115 6 132 29 197 36 174 1 178
170 125 177 158 145 28 149 37 116 147 42
198 121 96 120 72 86 46 135 76 144 162 192
54 159 104 185 79 89 83 57 190 194 70 187
169 81 105 94 61 139
```

Random / greedy / node exchange

min=58625 max=71791 avg=63910.23

worst = 71791



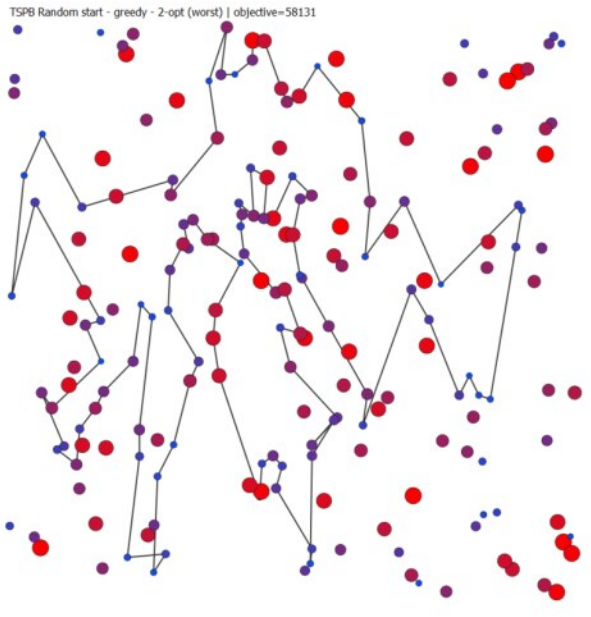
Worst solution nodes:

```
26 125 177 170 67 153 14 2 20 74 171 99 24
111 133 121 198 96 48 72 46 184 173 115 6
103 178 132 29 40 80 129 127 162 135 38
140 154 97 60 45 82 87 98 151 122 160 69
189 21 176 30 183 59 73 167 66 84 192 54
70 187 196 114 10 90 182 88 179 51 85 191
130 120 156 89 32 166 172 33 155 71 175
134 76 104 194 190 57 86 117 8 136 3 27 34
152 181 64 47 26
```

Random / greedy / 2-opt

min=51505 max=58131 avg=53797.47

worst = 58131



Worst solution nodes:

```
196 114 90 129 127 80 192 162 67 66 27 3
184 137 46 86 72 95 117 92 120 8 133 136
172 160 122 173 111 151 98 24 152 34 181
87 179 51 167 73 59 85 49 4 124 69 189 183
30 21 176 38 47 64 33 40 60 134 45 175 71
155 140 154 97 115 180 6 197 132 29 178
103 170 177 125 145 65 28 149 158 130 191
63 121 58 198 96 48 16 164 43 156 32 89 57
185 76 104 54 196
```

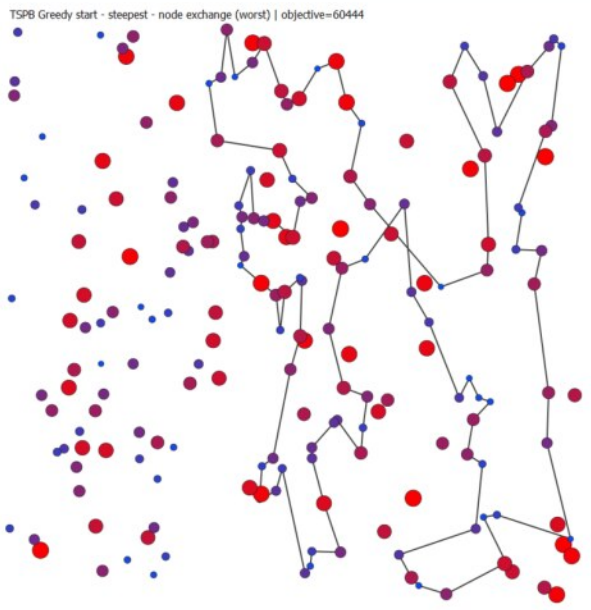
7. Worst solutions - TSPB

The plots show node cost using colour and size. The line represents the closed cycle.

Best greedy / steepest / node exchan

min=49785 max=60444 avg=53902.86

worst = 60444



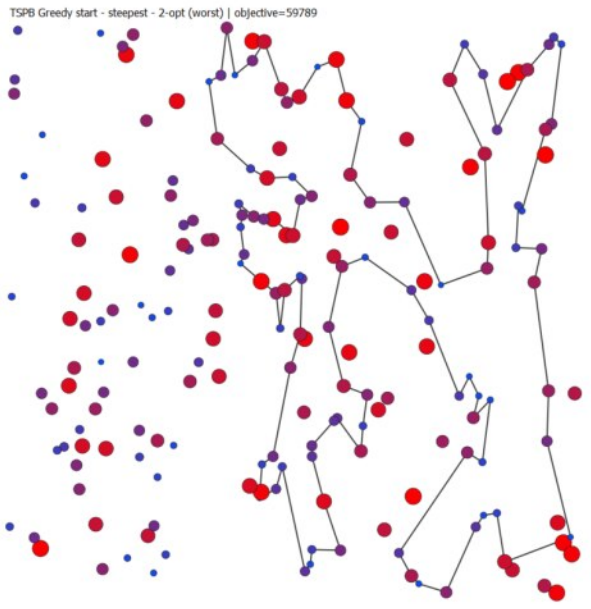
Worst solution nodes:

```
18 41 78 54 185 79 57 83 44 89 32 156 43
164 48 96 58 198 121 63 166 72 86 46 137
95 117 8 120 133 136 172 160 69 128 122
151 173 184 3 111 98 73 59 85 51 167 82 87
179 88 126 181 152 34 24 31 67 66 11 27
135 76 104 162 192 80 127 129 90 52 182
153 74 99 39 171 17 188 2 20 14 110 118 56
143 148 10 114 196 139 61 105 94 81 169
187 70 190 194 18
```

Best greedy / steepest / 2-o

min=49676 max=59789 avg=53306.10

worst = 59789



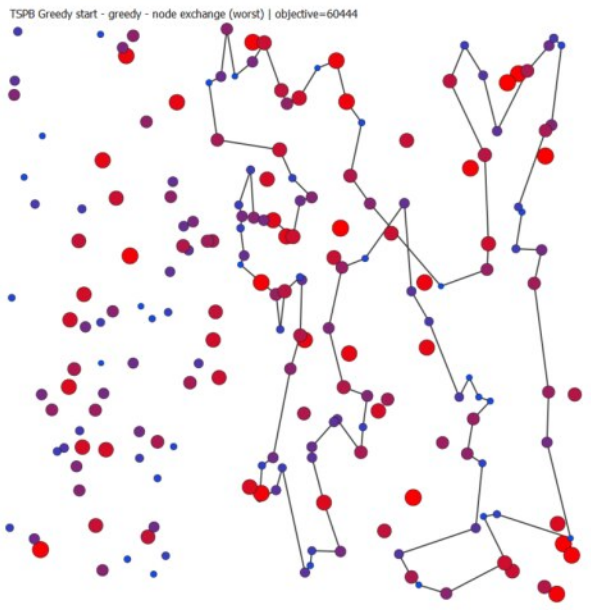
Worst solution nodes:

```
18 41 78 54 104 185 79 57 83 44 89 32 156
43 164 48 96 58 198 121 63 120 92 72 86 46
137 95 117 8 133 136 172 160 69 128 122
151 173 184 3 111 98 73 59 85 51 167 82 87
179 88 126 181 152 34 24 31 67 66 11 27
135 76 162 192 80 127 129 52 90 153 182 99
39 171 17 74 20 14 188 2 110 118 56 143
148 10 196 114 139 61 105 94 81 169 187 70
190 194 18
```

Best greedy / greedy / node exchang

min=49849 max=60444 avg=53922.85

worst = 60444



Worst solution nodes:

```
18 41 78 54 185 79 57 83 44 89 32 156 43
164 48 96 58 198 121 63 166 72 86 46 137
95 117 8 120 133 136 172 160 69 128 122
151 173 184 3 111 98 73 59 85 51 167 82 87
179 88 126 181 152 34 24 31 67 66 11 27
135 76 104 162 192 80 127 129 90 52 182
153 74 99 39 171 17 188 2 20 14 110 118 56
143 148 10 114 196 139 61 105 94 81 169
187 70 190 194 18
```

Best greedy / greedy / 2-opt

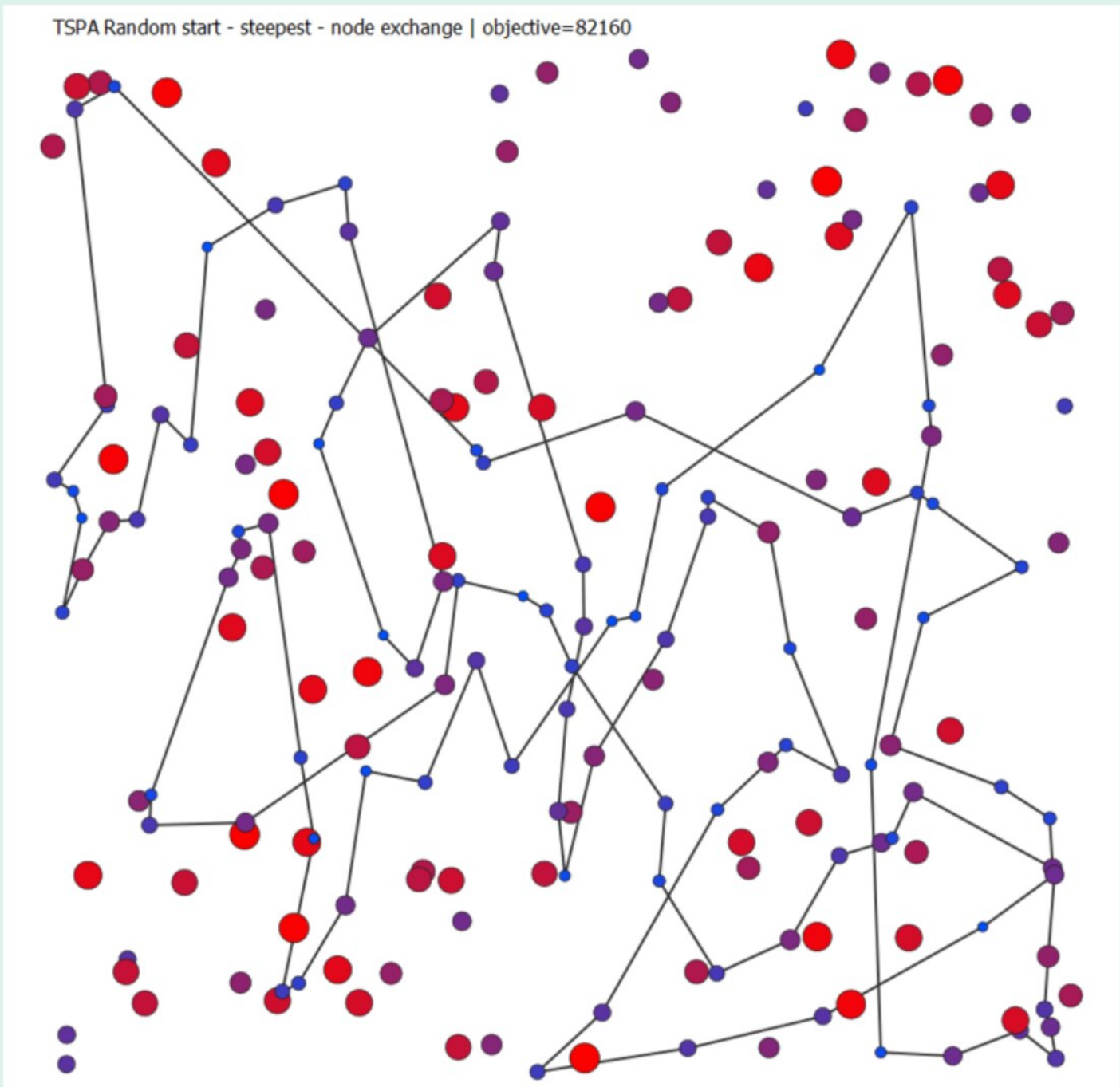
min=49710 max=59864 avg=53373.11

worst = 59864

Worst solution nodes:

```
18 41 78 54 104 185 79 57 83 44 89 32 156
43 164 48 96 58 198 121 63 133 120 92 72
86 46 137 95 117 8 136 172 160 69 128 122
151 173 184 3 111 98 73 59 85 51 167 82 87
179 88 126 181 152 34 24 31 67 66 11 27
135 76 162 192 80 127 129 90 52 182 153 14
20 74 99 39 171 17 188 2 110 118 56 143
148 10 196 114 139 61 105 94 81 169 187 70
190 194 18
```

8. Large best visualizations - TSPA



BEST SOLUTION

Random / steepest / node exchange

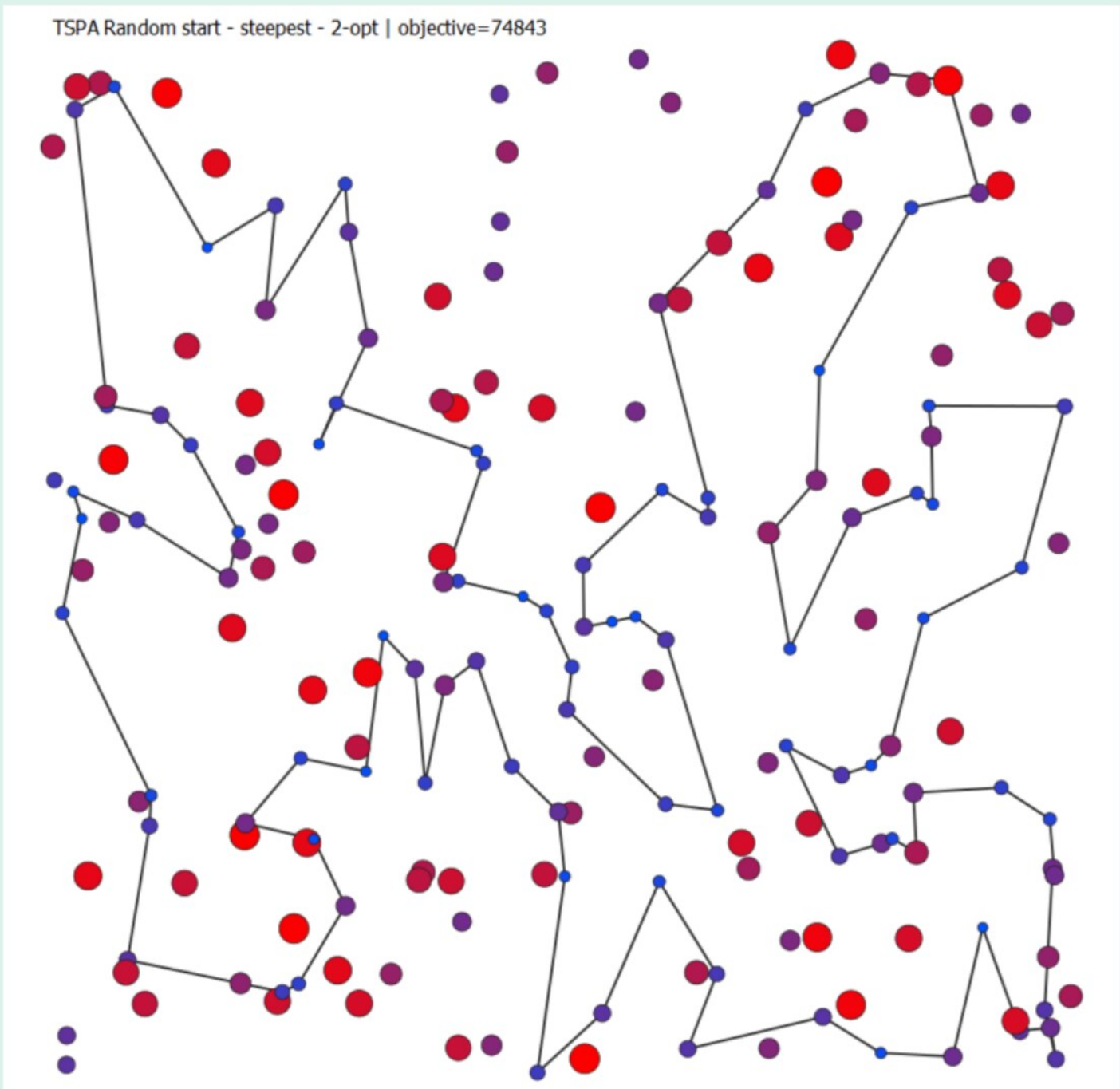
min=82160
max=98199
avg=90222.24

objective 82160

Best solution nodes:

```
107 182 192 114 181 134
176 150 174 8 9 54 131 41
135 73 140 179 52 139 112
4 76 132 180 28 156 71
157 91 80 152 108 49 11
119 168 63 149 116 60 31
48 51 16 143 85 3 50 47
93 43 130 159 146 113 89
42 151 138 158 58 84 53
144 67 35 102 161 127 97
160 83 123 90 30 187 171
2 189 141 87 185 45 177
36 33 164 178 1 169 106
57 64 88 78 193 153 75
170 107
```

100 selected nodes; closed route



BEST SOLUTION

Random / steepest / 2-opt

min=74843
max=80029
avg=77283.76

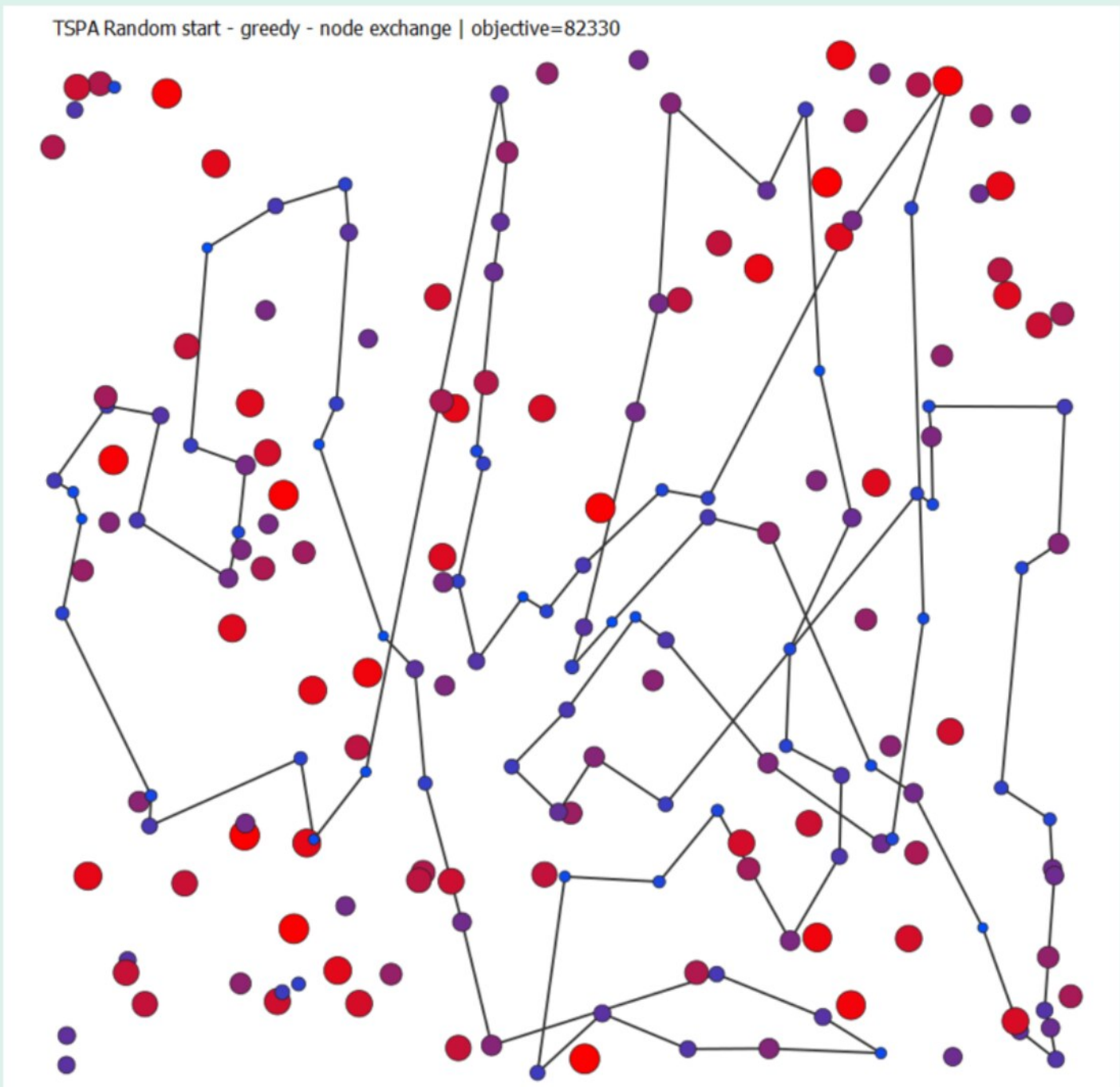
objective 74843

Best solution nodes:

```
135 114 181 58 158 192
182 57 64 107 88 193 78
153 146 170 159 130 113
103 89 42 151 176 150 106
43 93 47 147 1 169 50 3
85 174 8 142 164 178 101
29 163 136 186 62 9 54 33
52 179 177 36 131 134 84
140 53 144 67 35 156 16
51 76 132 4 71 157 38 91
80 48 31 60 108 152 123
160 49 149 63 168 97 127
69 171 2 189 187 161 30
141 180 28 87 102 185 45
73 135
```

100 selected nodes; closed route

8. Large best visualizations - TSPA



BEST SOLUTION

Random / greedy / node exchange

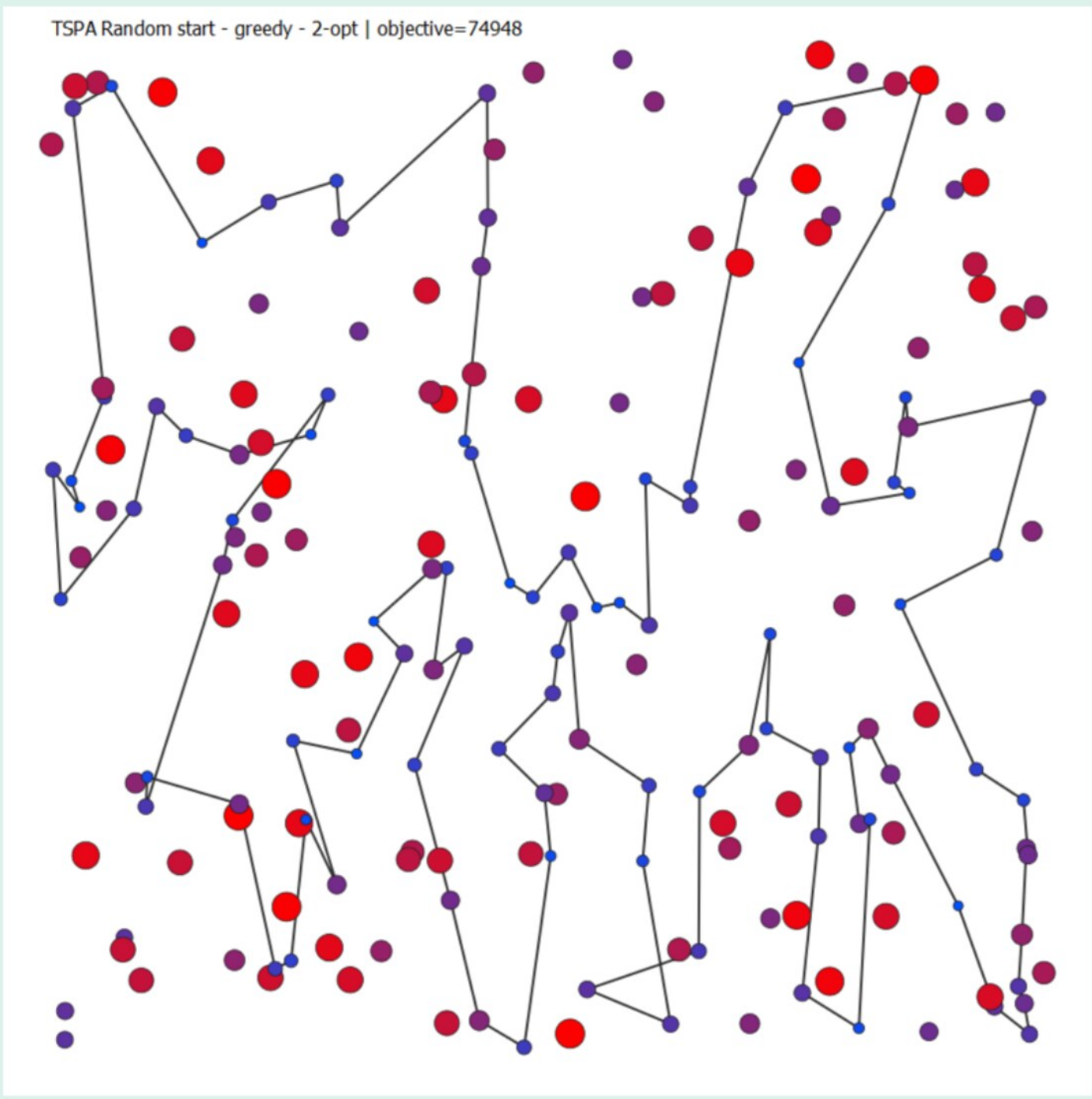
min=82330
max=95592
avg=87940.23

objective 82330

Best solution nodes:

89 93 178 29 184 9 33 52
144 67 185 35 16 51 139
112 5 115 141 187 30 127
97 168 63 149 116 60 108
49 160 123 100 152 80 91
157 71 76 132 180 28 87
98 25 158 182 57 111 192
181 114 135 58 134 138
151 150 176 174 85 164
136 186 154 62 143 179 53
177 54 194 8 106 113 107
88 78 193 153 170 146 159
130 47 37 147 1 169 50 3
84 41 73 45 140 36 131 72
42 89

100 selected nodes; closed route



BEST SOLUTION

Random / greedy / 2-opt

min=74948
max=80421
avg=77398.37

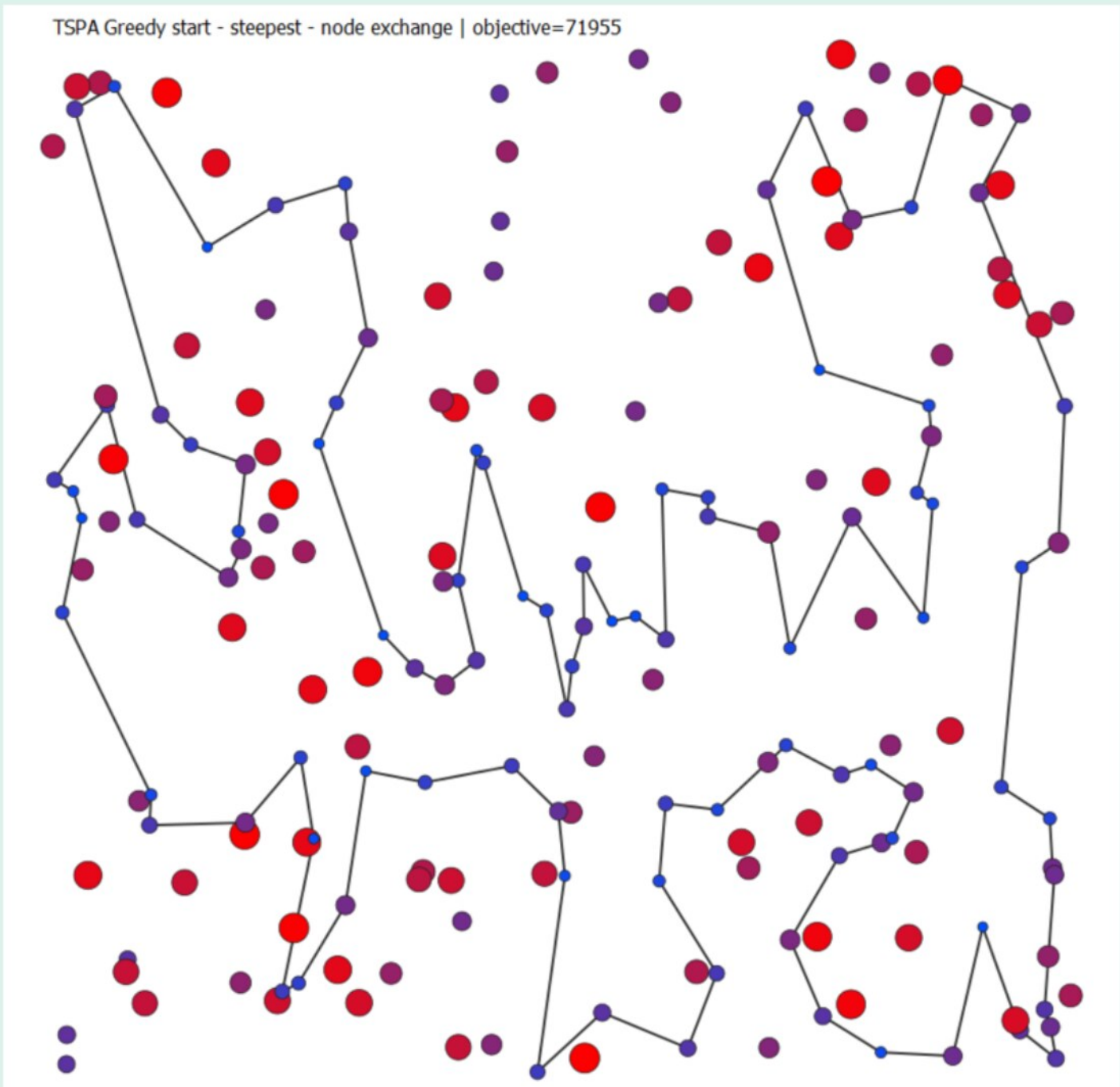
objective 74948

Best solution nodes:

78 193 153 75 146 170 159
130 93 47 147 169 1 3 50
85 164 178 29 136 186 9
54 33 131 36 177 52 144
67 16 51 139 112 115 71
157 91 80 48 31 60 149 63
116 168 49 108 152 100
132 76 123 160 127 97 161
171 2 187 189 30 141 28
180 156 35 102 185 87 98
25 114 135 73 45 140 53
179 41 84 58 192 181 158
134 72 174 176 150 151
182 57 89 42 106 43 113
107 88 78

100 selected nodes; closed route

8. Large best visualizations - TSPA



BEST SOLUTION

Best greedy / steepest / node exchange

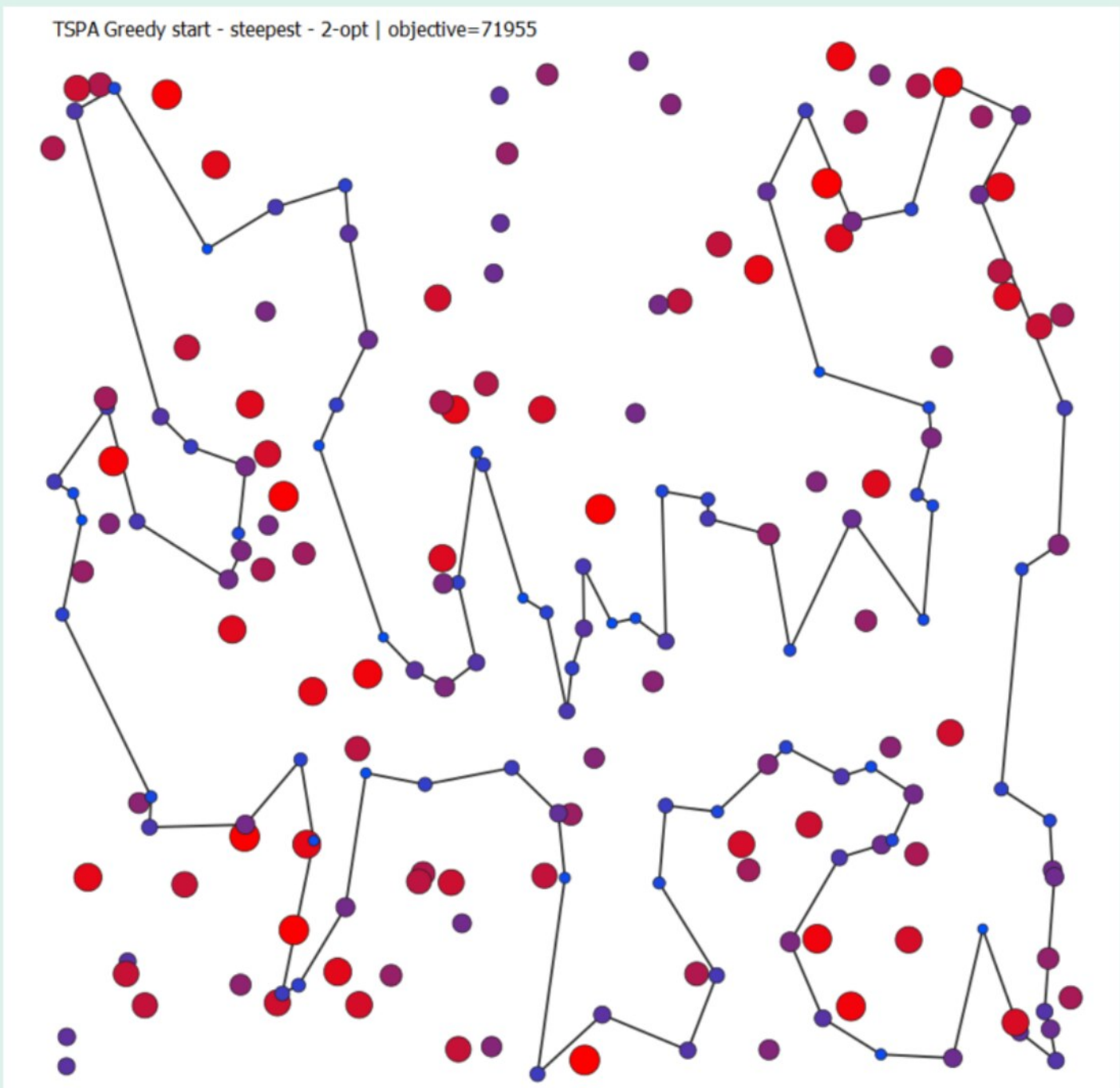
min=71955
max=74861
avg=72983.04

objective 71955

Best solution nodes:

101 55 29 178 184 136 186
164 1 169 3 50 93 85 174
8 54 9 33 131 36 177 52
179 53 140 144 67 16 51
35 185 102 28 180 132 76
4 71 157 91 80 48 31 108
152 100 123 83 160 49 60
116 149 63 168 97 127 161
30 187 171 2 189 141 87
45 73 135 114 181 192 158
58 84 134 72 176 150 106
113 89 42 151 138 182 57
64 107 88 78 193 153 170
146 159 130 47 37 147 101

100 selected nodes; closed route



BEST SOLUTION

Best greedy / steepest / 2-opt

min=71955
max=74700
avg=73003.71

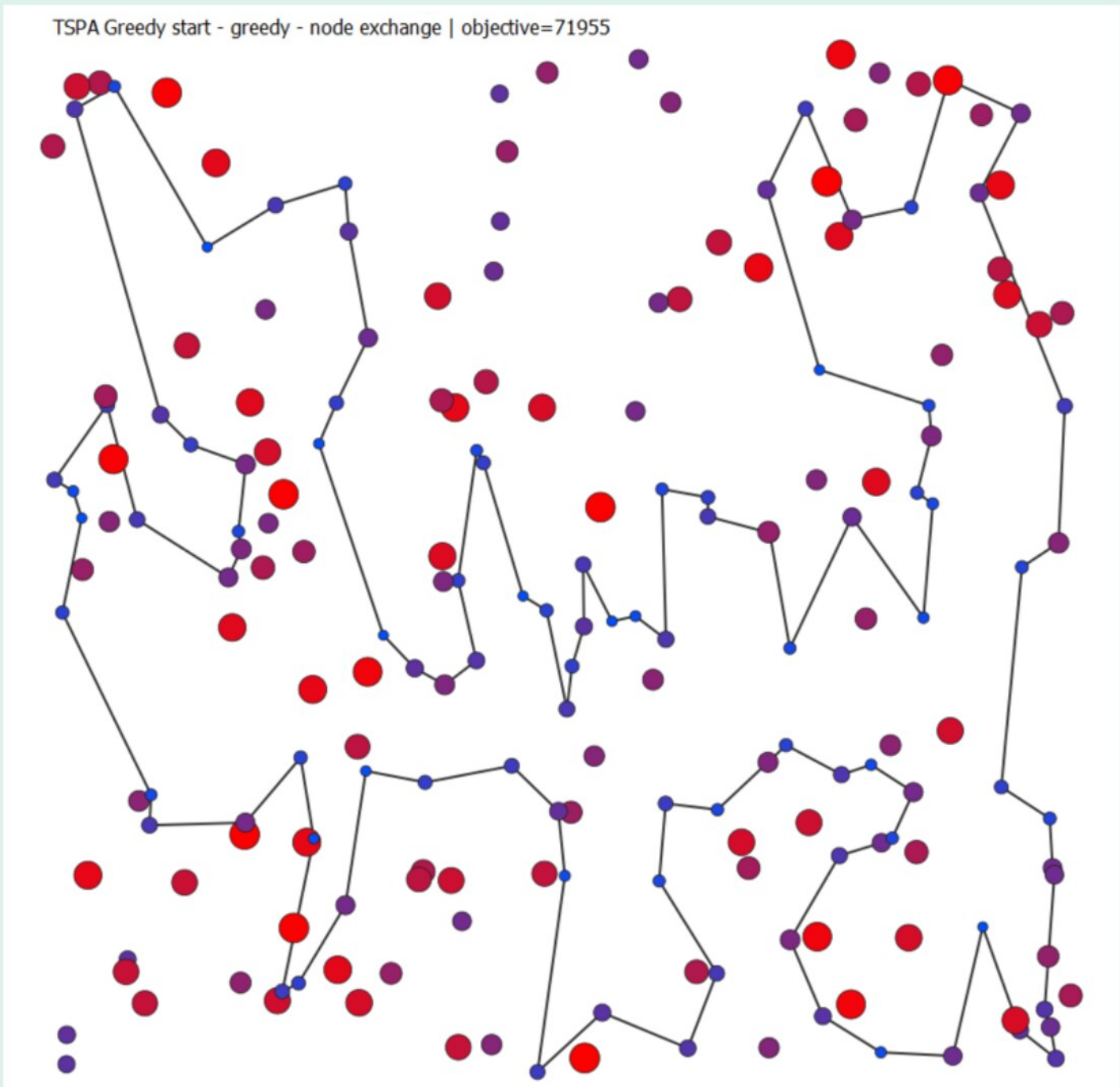
objective 71955

Best solution nodes:

55 101 147 37 47 130 159
146 170 153 193 78 88 107
64 57 182 138 151 42 89
113 106 150 176 72 134 84
58 158 192 181 114 135 73
45 87 141 189 2 171 187
30 161 127 97 168 63 149
116 60 49 160 83 123 100
152 108 31 48 80 91 157
71 4 76 132 180 28 102
185 35 51 16 67 144 140
53 179 52 177 36 131 33 9
54 8 174 85 93 50 3 169 1
164 186 136 184 178 29 55

100 selected nodes; closed route

8. Large best visualizations - TSPA



BEST SOLUTION

Best greedy / greedy / node exchange

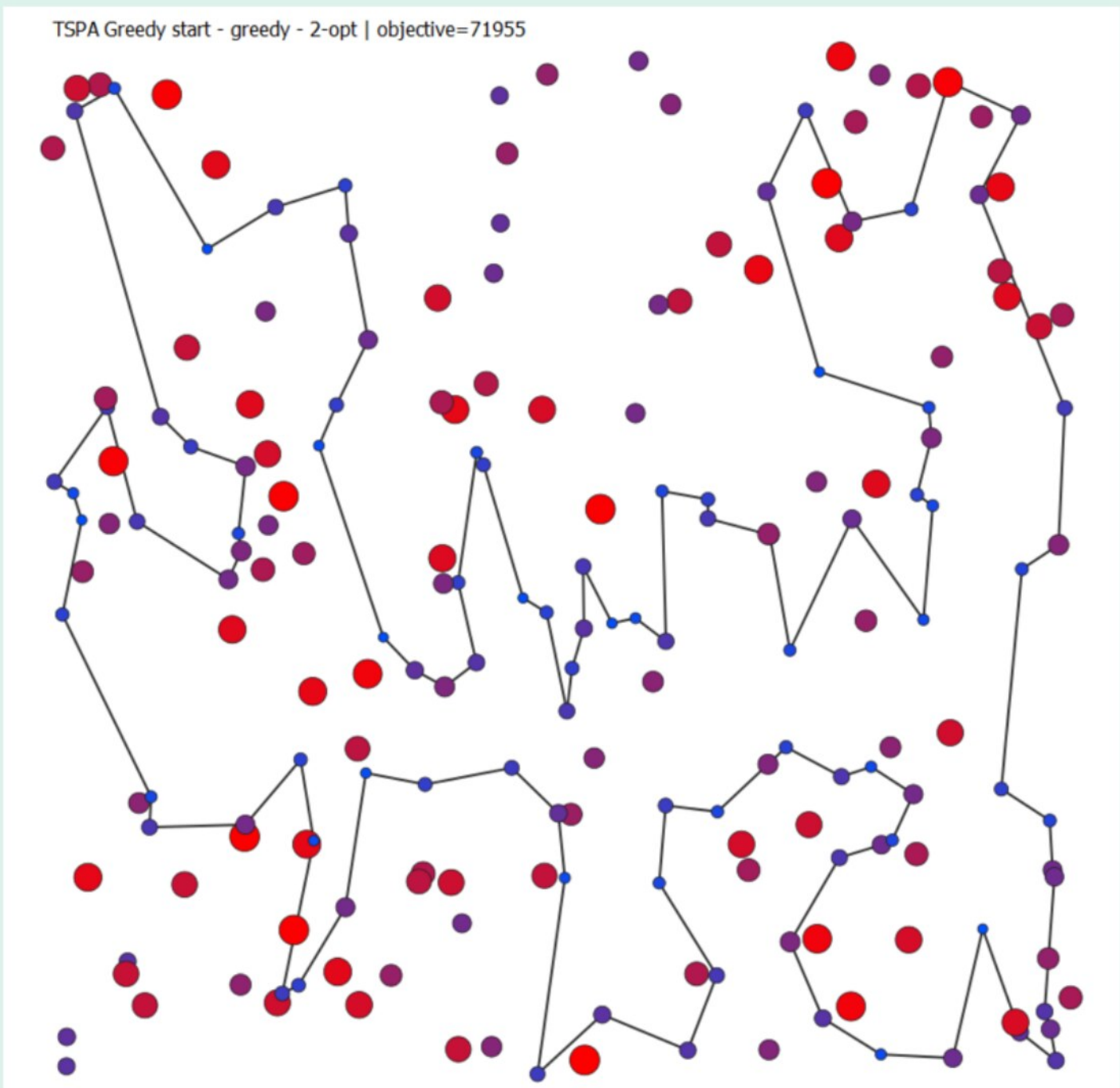
min=71955
max=74861
avg=72984.80

objective 71955

Best solution nodes:

```
101 55 29 178 184 136 186
164 1 169 3 50 93 85 174
8 54 9 33 131 36 177 52
179 53 140 144 67 16 51
35 185 102 28 180 132 76
4 71 157 91 80 48 31 108
152 100 123 83 160 49 60
116 149 63 168 97 127 161
30 187 171 2 189 141 87
45 73 135 114 181 192 158
58 84 134 72 176 150 106
113 89 42 151 138 182 57
64 107 88 78 193 153 170
146 159 130 47 37 147 101
```

100 selected nodes; closed route



BEST SOLUTION

Best greedy / greedy / 2-opt

min=71955
max=74844
avg=73015.66

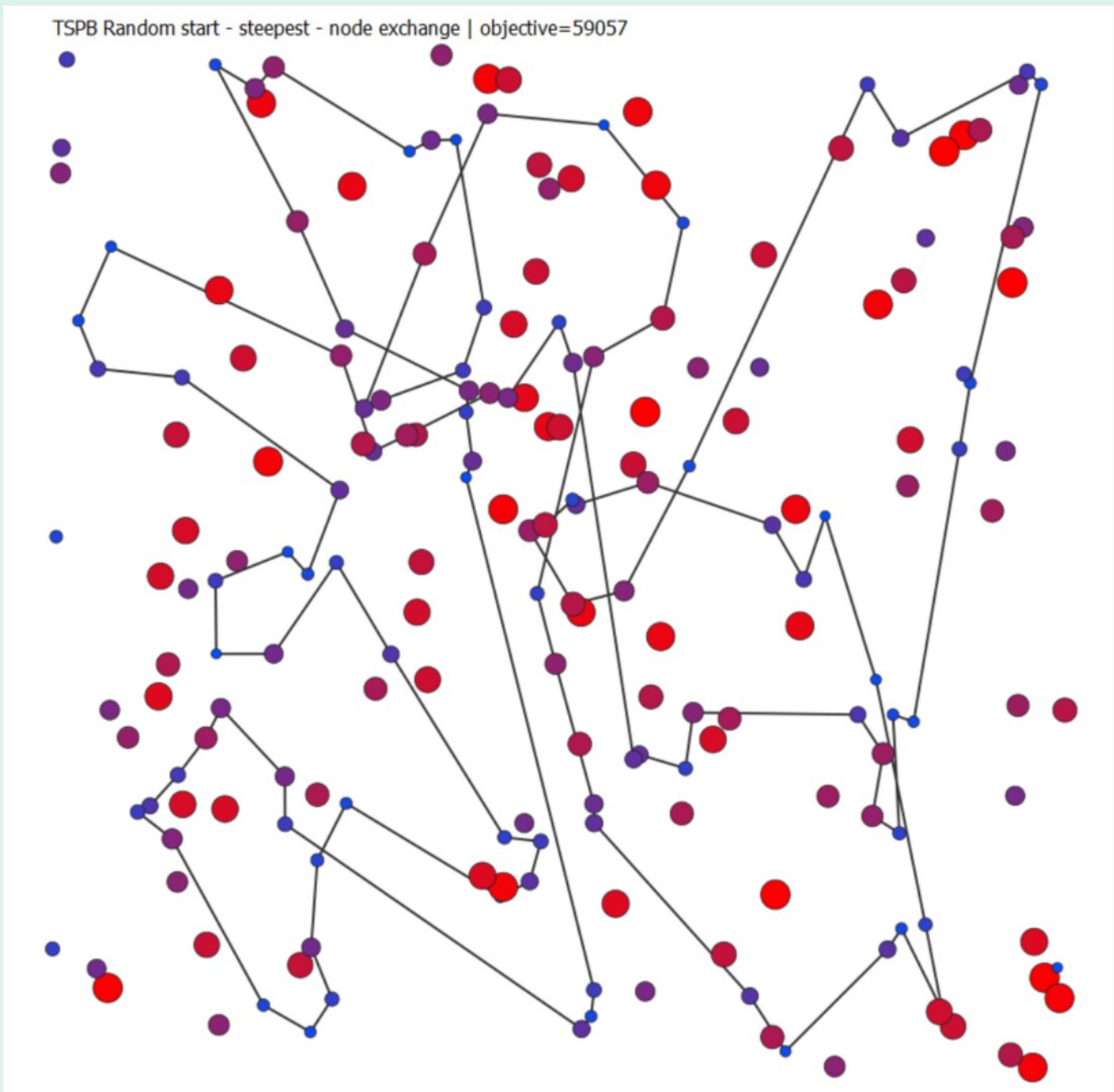
objective 71955

Best solution nodes:

```
55 101 147 37 47 130 159
146 170 153 193 78 88 107
64 57 182 138 151 42 89
113 106 150 176 72 134 84
58 158 192 181 114 135 73
45 87 141 189 2 171 187
30 161 127 97 168 63 149
116 60 49 160 83 123 100
152 108 31 48 80 91 157
71 4 76 132 180 28 102
185 35 51 16 67 144 140
53 179 52 177 36 131 33 9
54 8 174 85 93 50 3 169 1
164 186 136 184 178 29 55
```

100 selected nodes; closed route

8. Large best visualizations - TSPB



BEST SOLUTION

Random / steepest / node exchange

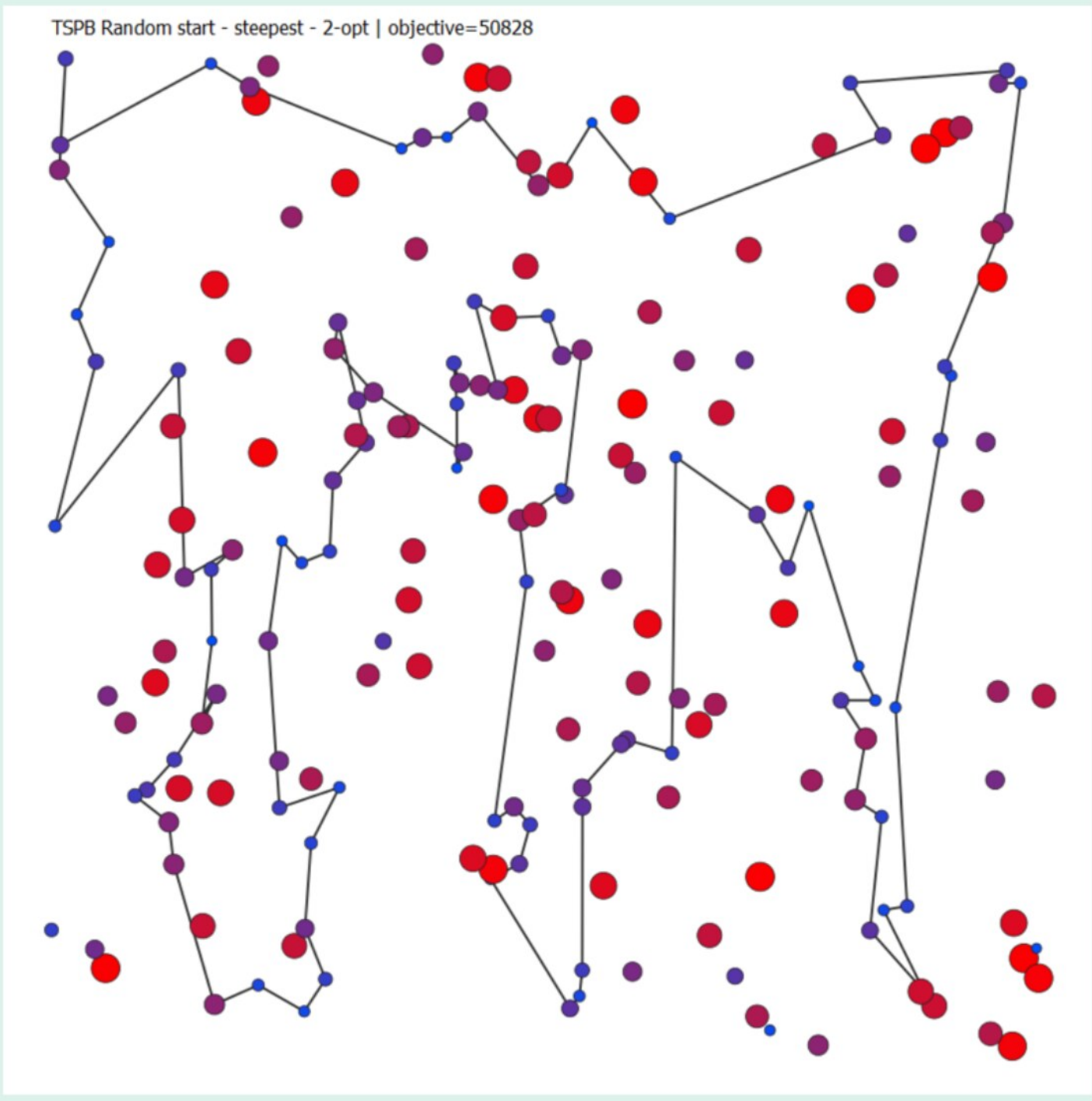
min=59057
max=73129
avg=65886.03

objective 59057

Best solution nodes:

```
180 6 29 132 197 175 134
45 157 60 40 85 51 167 59
64 47 97 170 125 154 140
38 158 145 28 149 191 21
189 8 117 72 46 34 24 67
66 80 52 182 153 129 90
10 114 105 94 70 190 76
27 111 122 173 184 3 135
162 192 54 127 14 2 20 74
171 99 181 152 98 151 86
79 57 89 48 63 30 183 133
120 96 198 121 42 147 116
50 130 136 172 160 69 179
87 82 71 155 115 180
```

100 selected nodes; closed route



BEST SOLUTION

Random / steepest / 2-opt

min=50828
max=56237
avg=53457.16

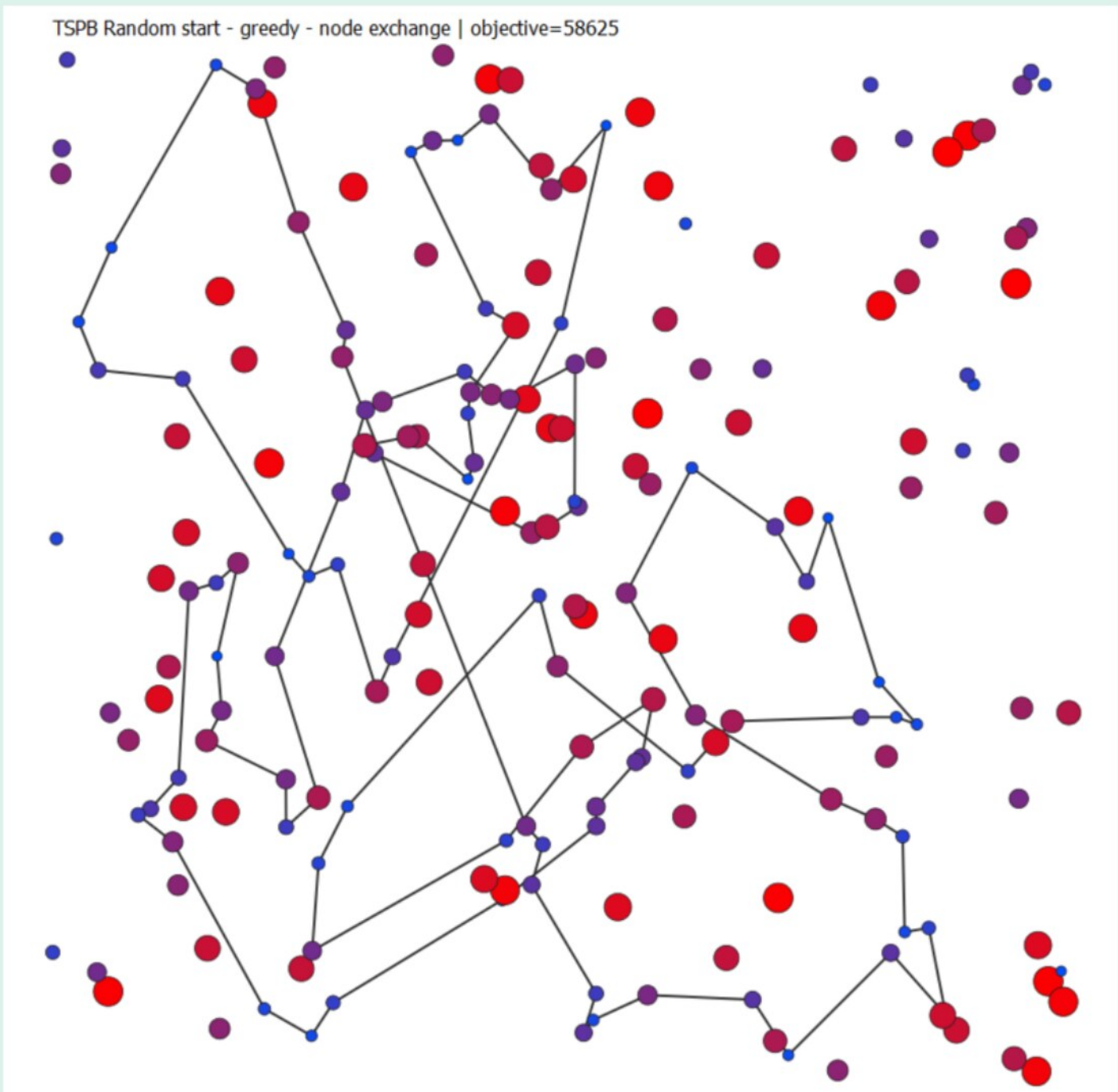
objective 50828

Best solution nodes:

```
196 61 105 81 94 190 70
57 89 32 156 48 96 198
121 147 116 146 37 13 149
28 145 65 158 177 26 125
170 180 115 6 29 132 197
36 62 175 134 45 157 60
40 71 97 154 140 47 38 21
30 130 191 183 160 69 172
133 136 8 117 120 92 72
46 86 3 184 122 151 59 73
167 51 85 82 87 179 181
152 34 24 67 76 162 192
54 127 129 80 52 182 153
74 2 20 14 90 10 114 196
```

100 selected nodes; closed route

8. Large best visualizations - TSPB



BEST SOLUTION

Random / greedy / node exchange

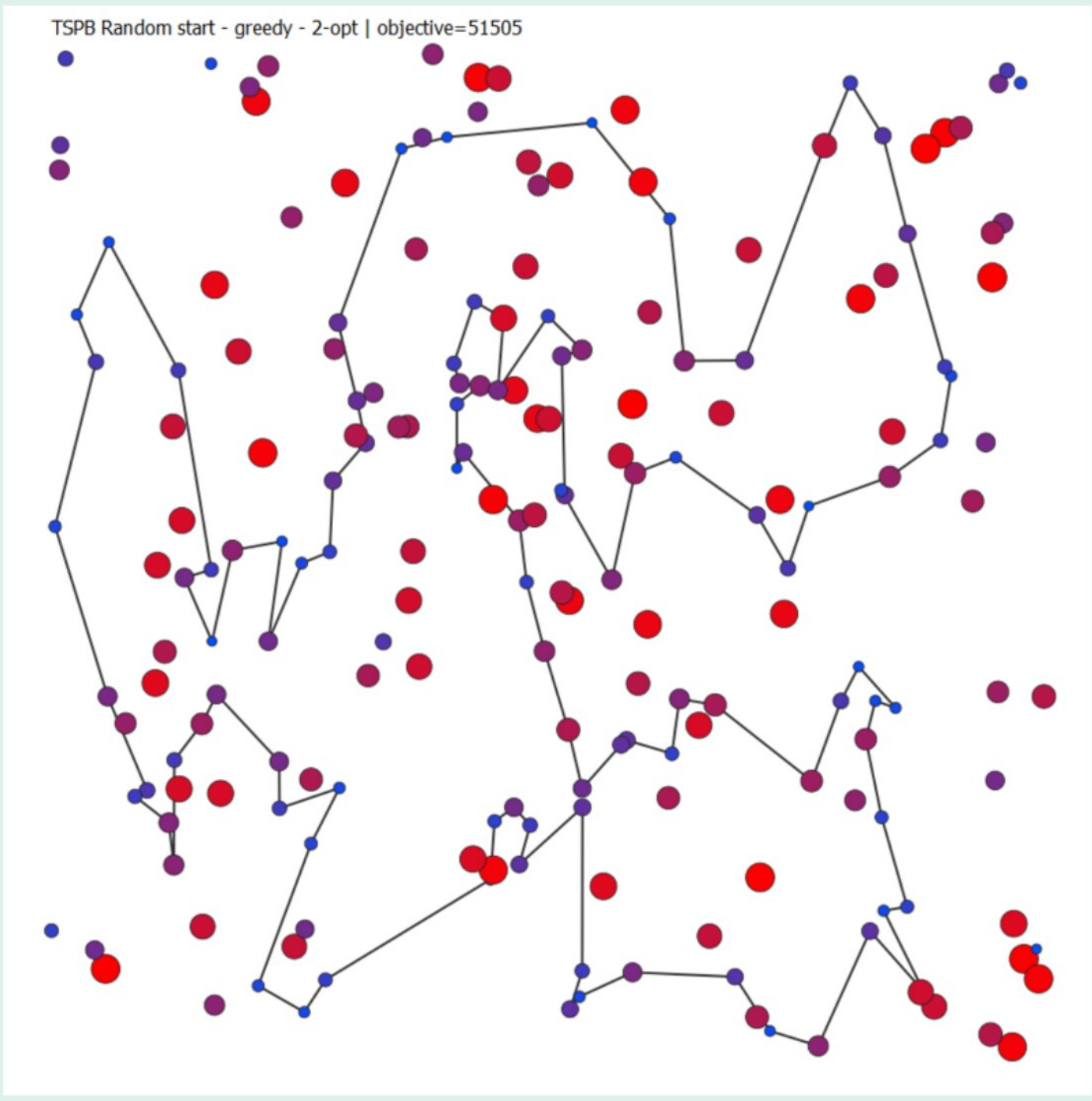
min=58625
max=71791
avg=63910.23

objective 58625

Best solution nodes:

```
155 71 141 97 38 30 183
133 8 117 46 184 3 173
122 21 176 189 168 69 160
172 136 92 120 121 198 96
48 156 89 72 64 33 47 140
154 158 145 28 149 116
147 50 130 191 73 167 51
179 82 87 88 99 171 74 2
14 20 153 182 150 66 27
76 162 192 54 127 90 129
80 84 67 98 151 40 60 157
59 25 11 24 34 152 181 85
45 134 175 197 132 29 6
177 125 26 170 115 180
155
```

100 selected nodes; closed route



BEST SOLUTION

Random / greedy / 2-opt

min=51505
max=58131
avg=53797.47

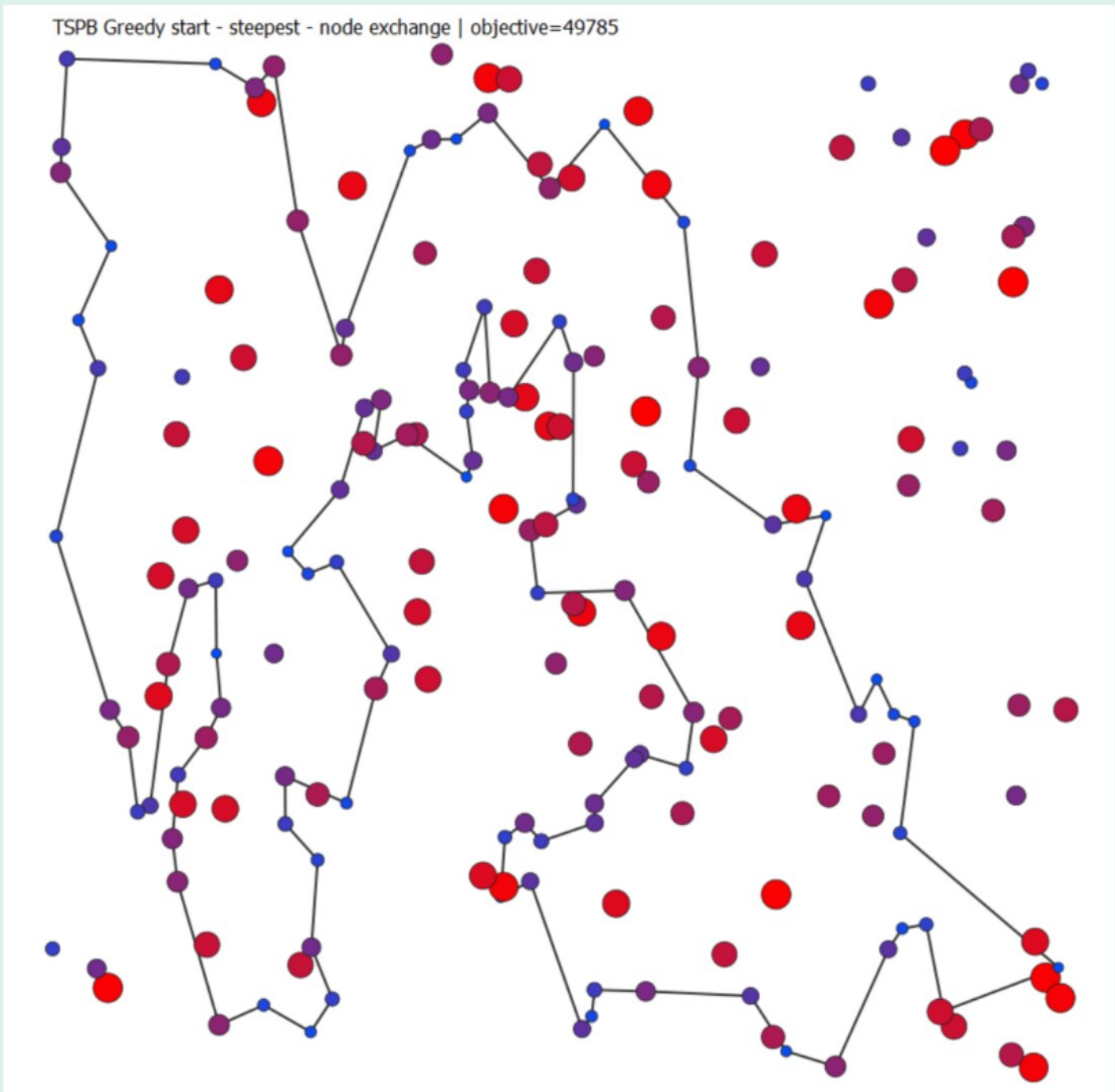
objective 51505

Best solution nodes:

```
34 24 67 66 84 150 80 127
90 129 52 153 14 20 2 74
17 171 99 88 87 82 179
181 51 167 73 59 85 45
134 175 60 40 71 155 115
180 6 36 197 132 29 178
65 145 28 149 158 125 177
170 26 154 97 140 47 38
21 30 130 121 96 89 57
185 104 194 190 70 187
196 114 10 78 54 192 162
76 135 27 184 3 46 86 72
117 92 120 133 136 8 172
69 160 122 151 98 25 152
34
```

100 selected nodes; closed route

8. Large best visualizations - TSPB



BEST SOLUTION

Best greedy / steepest / node exchange

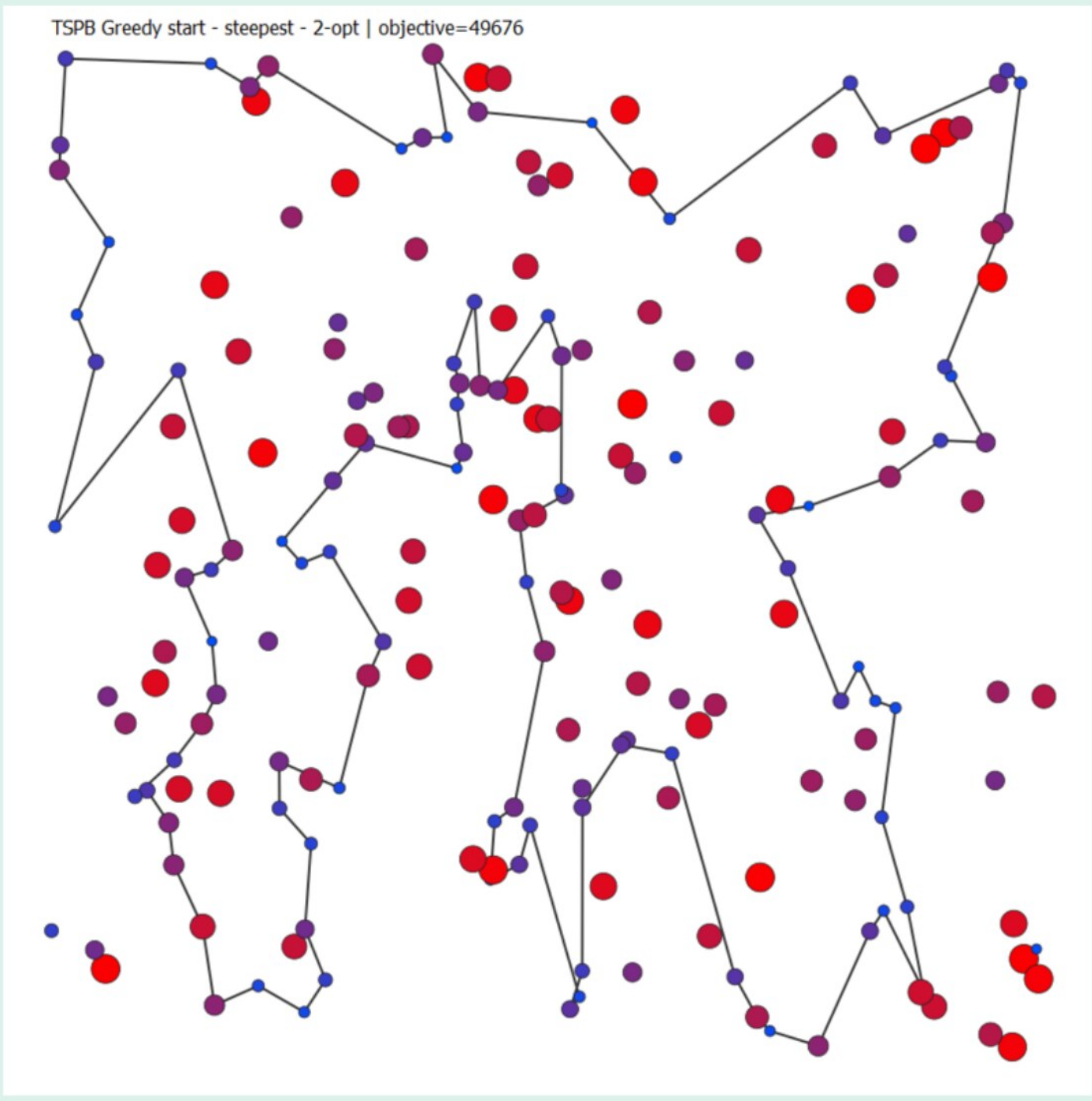
min=49785
max=60444
avg=53902.86

objective 49785

Best solution nodes:

110 2 14 20 74 17 171 99
88 179 87 82 51 85 59 73
167 181 152 34 24 67 66
27 151 122 3 184 46 72
117 8 120 133 136 172 160
69 189 21 183 30 38 154
140 47 64 33 40 155 71 60
157 45 134 175 62 36 197
6 180 115 170 125 177 19
29 132 103 178 65 145 28
149 13 146 37 116 147 42
50 191 130 121 198 96 48
156 89 57 185 76 162 54
192 80 127 129 90 153 110

100 selected nodes; closed route



BEST SOLUTION

Best greedy / steepest / 2-opt

min=49676
max=59789
avg=53306.10

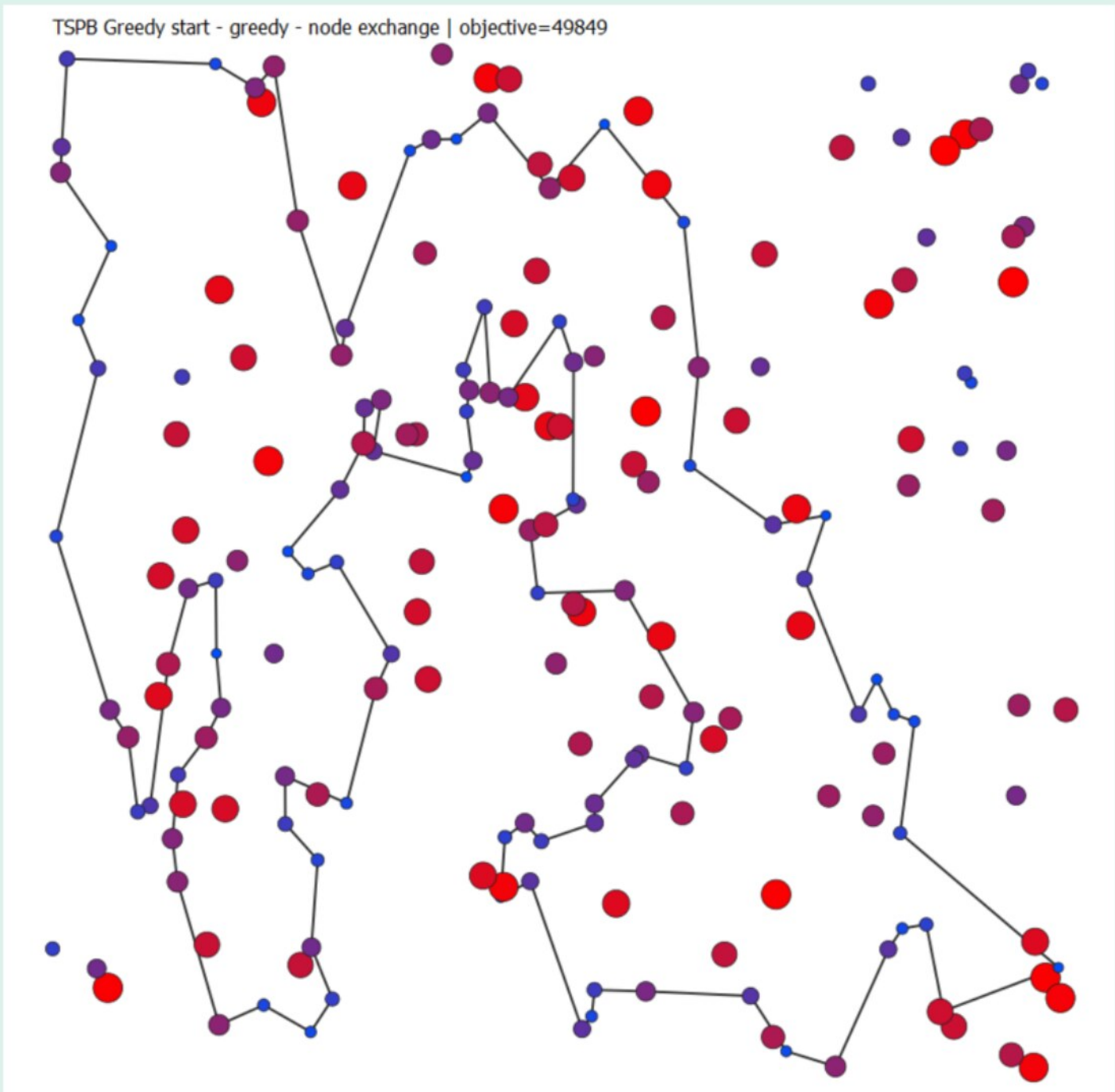
objective 49676

Best solution nodes:

61 196 114 148 10 78 54
162 192 80 127 129 90 153
14 2 20 74 17 171 39 99
67 34 24 181 179 82 87
167 51 85 59 73 98 151
122 3 184 46 72 117 8 120
133 136 172 160 69 21 38
154 140 47 64 33 40 155
71 60 157 45 134 175 62
91 36 197 29 132 6 180
115 170 177 125 26 158 65
145 28 149 13 146 37 116
147 42 121 198 96 58 48
89 57 190 70 81 94 105 61

100 selected nodes; closed route

8. Large best visualizations - TSPB



BEST SOLUTION

Best greedy / greedy / node exchange

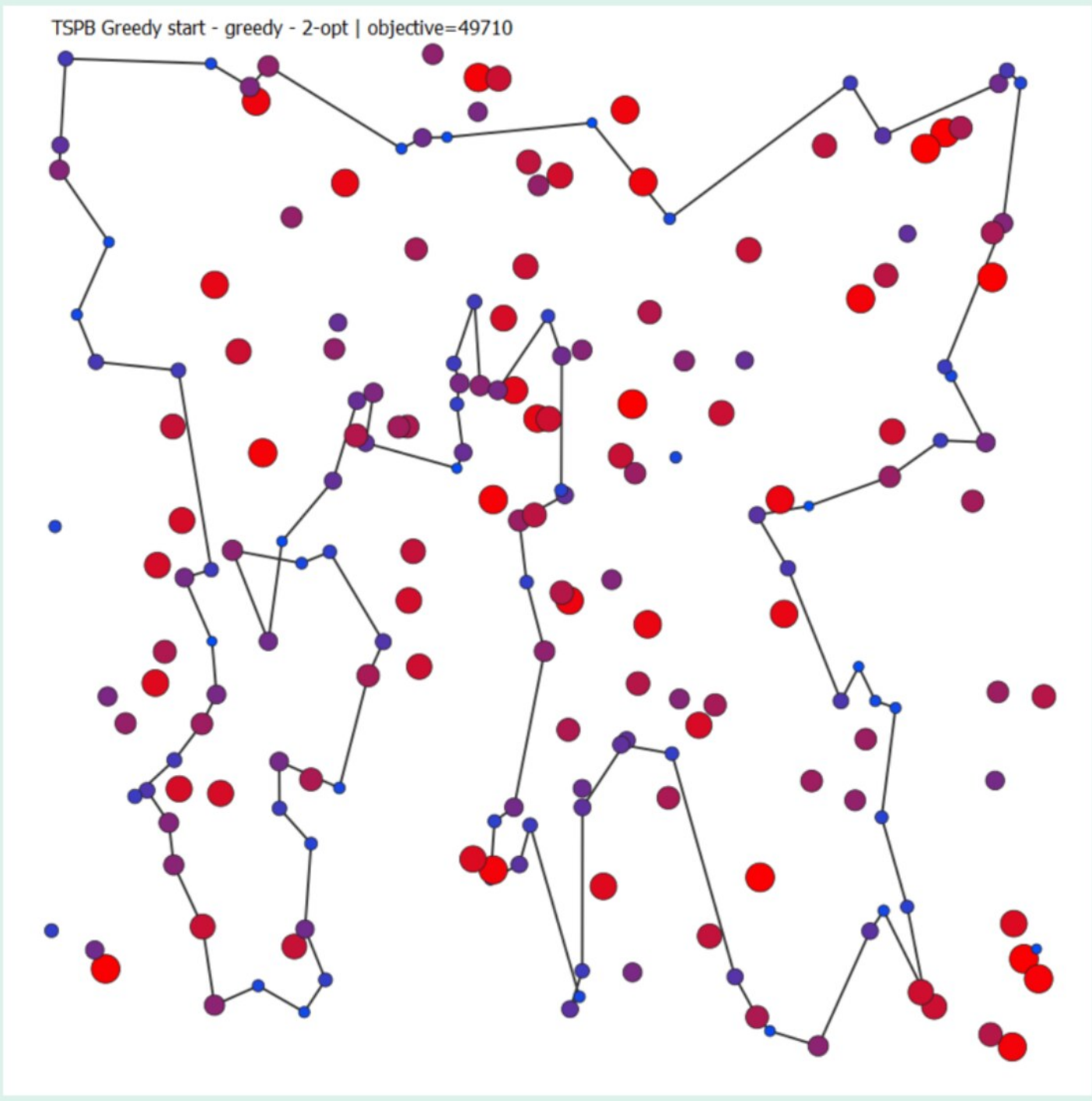
min=49849
max=60444
avg=53922.85

objective 49849

Best solution nodes:

```
110 2 14 20 74 17 171 99
88 179 87 82 51 85 59 73
167 181 152 34 24 67 66
27 151 122 3 184 46 72
117 8 120 133 136 172 160
69 21 183 30 176 38 154
140 47 64 33 40 155 71 60
157 45 134 175 62 36 197
6 180 115 170 125 177 19
29 132 103 178 65 145 28
149 13 146 37 116 147 42
50 191 130 121 198 96 48
156 89 57 185 76 162 54
192 80 127 129 90 153 110
```

100 selected nodes; closed route



BEST SOLUTION

Best greedy / greedy / 2-opt

min=49710
max=59864
avg=53373.11

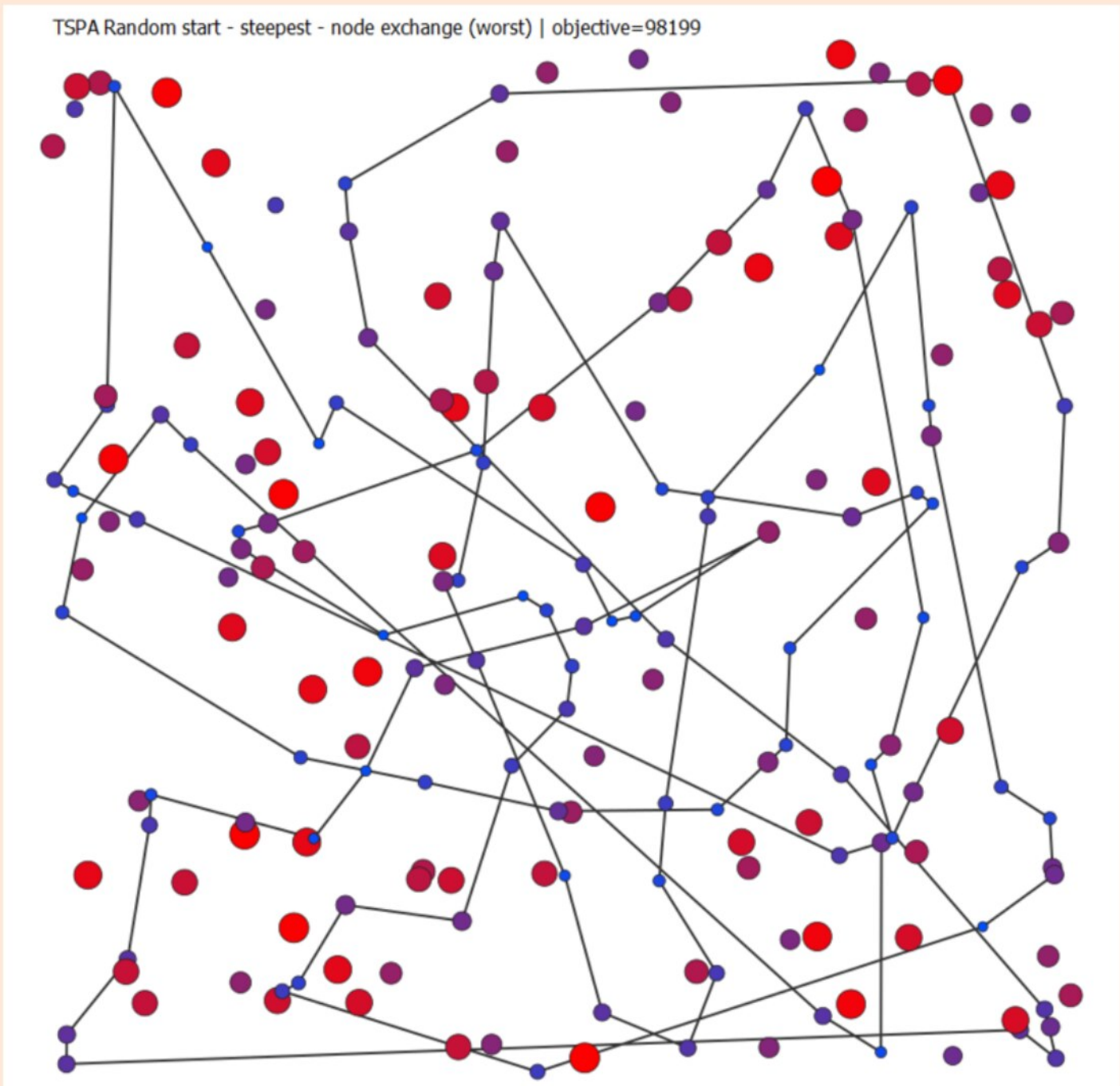
objective 49710

Best solution nodes:

```
38 154 97 26 140 47 64 33
40 155 71 60 157 45 134
175 62 91 36 197 29 132 6
180 115 170 177 125 158
145 28 149 13 146 37 116
147 42 121 198 96 89 57
190 70 81 94 105 61 196
114 148 10 78 54 162 192
80 127 129 90 153 14 2 20
74 17 171 39 99 67 34 24
181 179 82 87 167 51 85
59 73 98 151 122 3 184 46
72 117 8 120 133 136 172
160 69 21 183 30 38
```

100 selected nodes; closed route

8. Large worst visualizations - TSPA



WORST SOLUTION

Random / steepest / node exchange

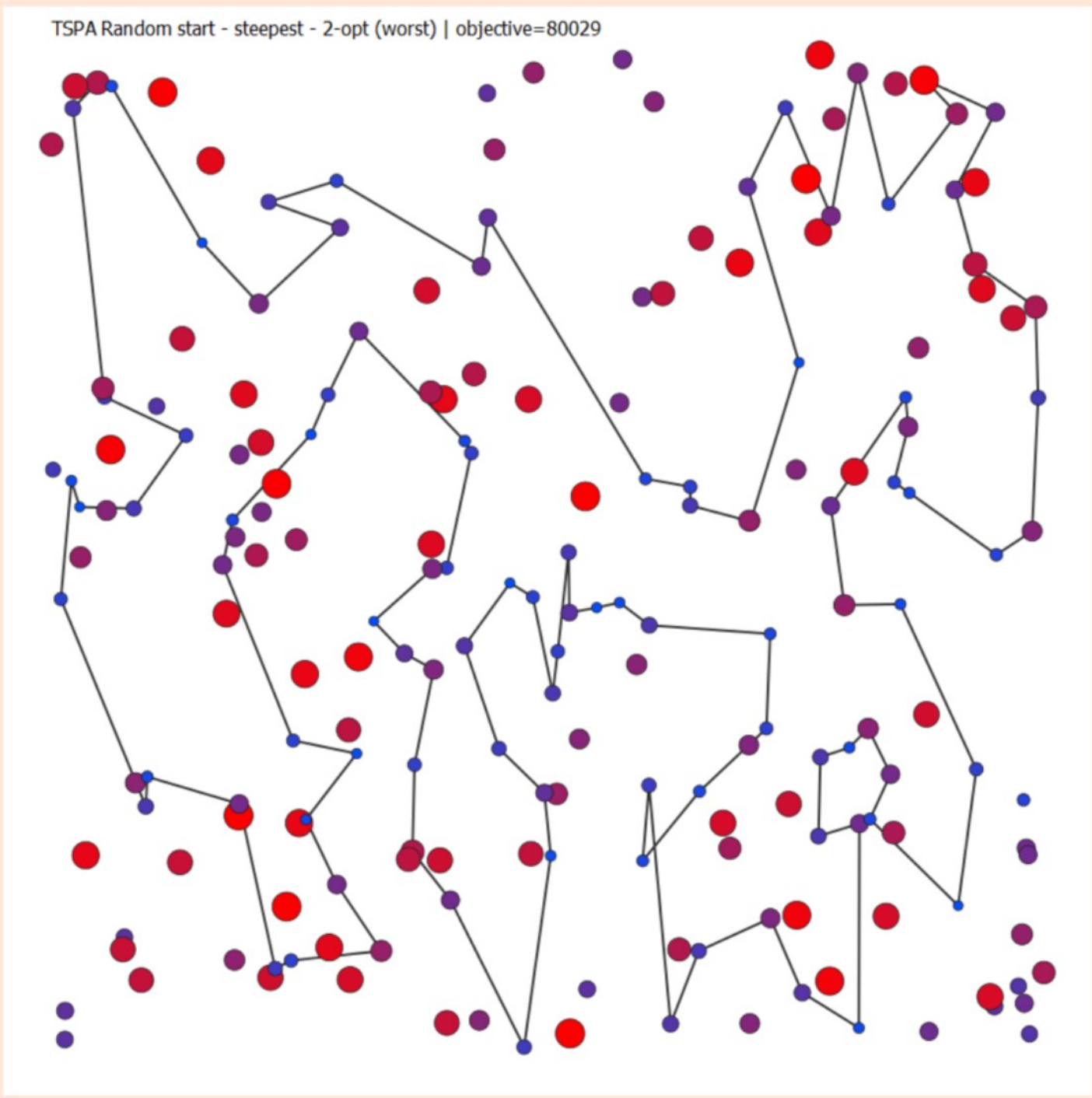
min=82160
max=98199
avg=90222.24

objective 98199

Worst solution nodes:

```
194 8 179 28 141 187 97
127 69 129 104 88 78 153
150 131 4 71 157 115 29
147 37 47 113 89 106 43
93 184 136 186 62 51 90
123 83 180 67 144 53 140
45 98 189 2 171 114 107
170 146 159 130 169 1 178
164 9 54 84 58 158 192
181 135 185 156 35 16 139
112 33 85 3 50 174 176
134 73 87 30 168 63 108
152 182 57 42 151 49 149
116 60 48 80 132 76 52
177 36 194
```

100 selected nodes; closed route



WORST SOLUTION

Random / steepest / 2-opt

min=74843
max=80029
avg=77283.76

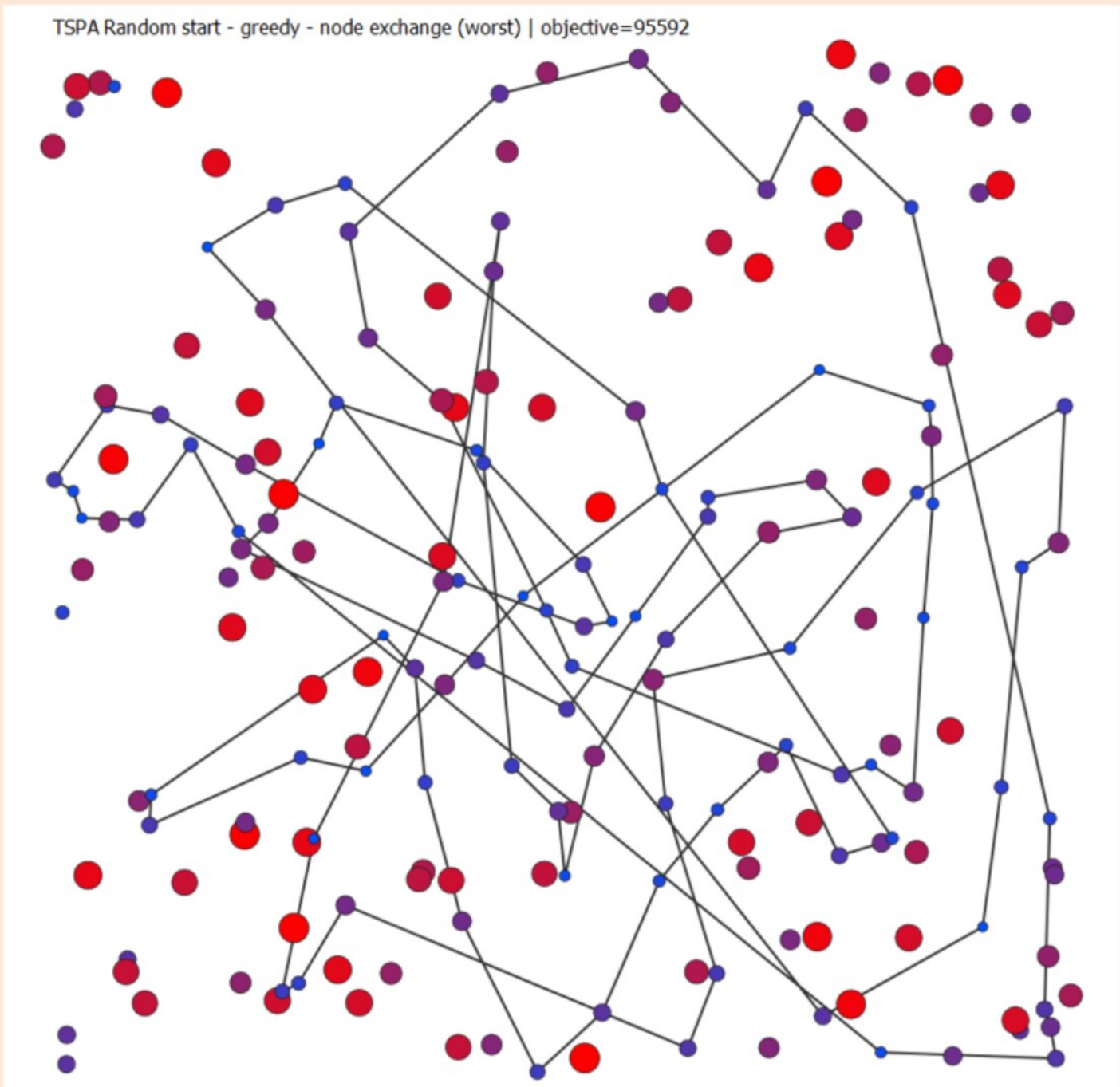
objective 80029

Worst solution nodes:

```
8 164 186 136 184 163 178
17 29 55 101 46 21 147 37
47 50 3 169 1 85 81 93
130 107 89 113 43 106 150
151 42 57 182 138 158 192
84 58 134 176 174 131 36
177 179 52 53 140 144 67
185 45 73 135 114 98 27
87 102 28 180 156 35 16
51 4 76 132 123 160 30
141 187 189 10 2 171 161
97 127 168 149 63 49 152
60 31 7 48 80 38 71 91
157 139 112 33 9 54 8
```

100 selected nodes; closed route

8. Large worst visualizations - TSPA



WORST SOLUTION

Random / greedy / node exchange

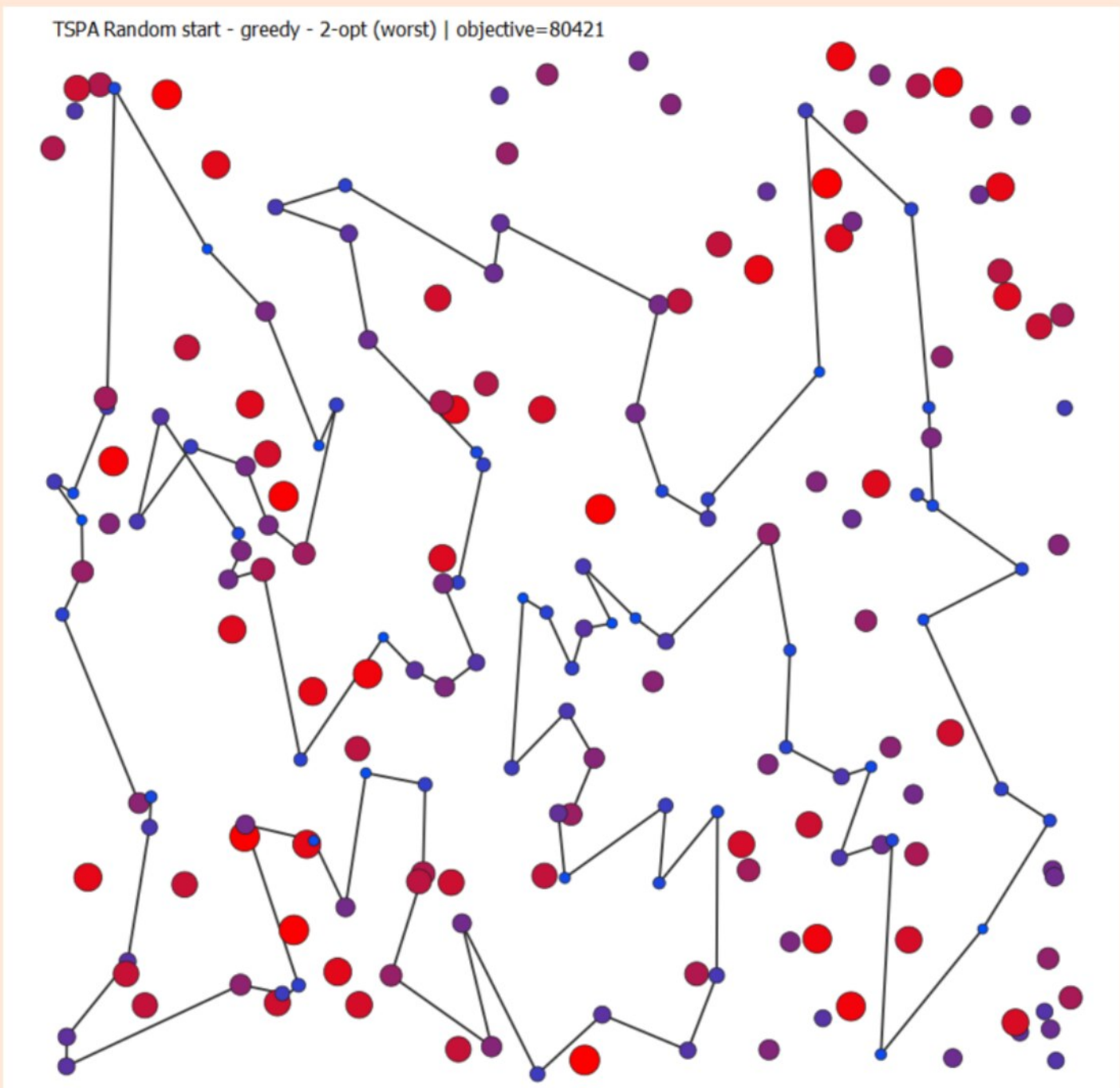
min=82330
max=95592
avg=87940.23

objective 95592

Worst solution nodes:

185 140 36 54 9 142 85 8
194 131 41 135 73 45 16
139 112 156 187 171 2 189
192 158 84 99 174 3 147
37 47 130 107 182 38 80
91 157 143 33 89 42 151
176 72 134 58 181 114 98
87 28 180 97 127 30 141
102 67 164 1 169 50 93
113 106 150 53 144 196 4
71 115 39 186 136 178 159
153 193 78 57 123 152 49
63 149 116 60 108 100 35
179 177 52 51 76 132 90
83 185

100 selected nodes; closed route



WORST SOLUTION

Random / greedy / 2-opt

min=74948
max=80421
avg=77398.37

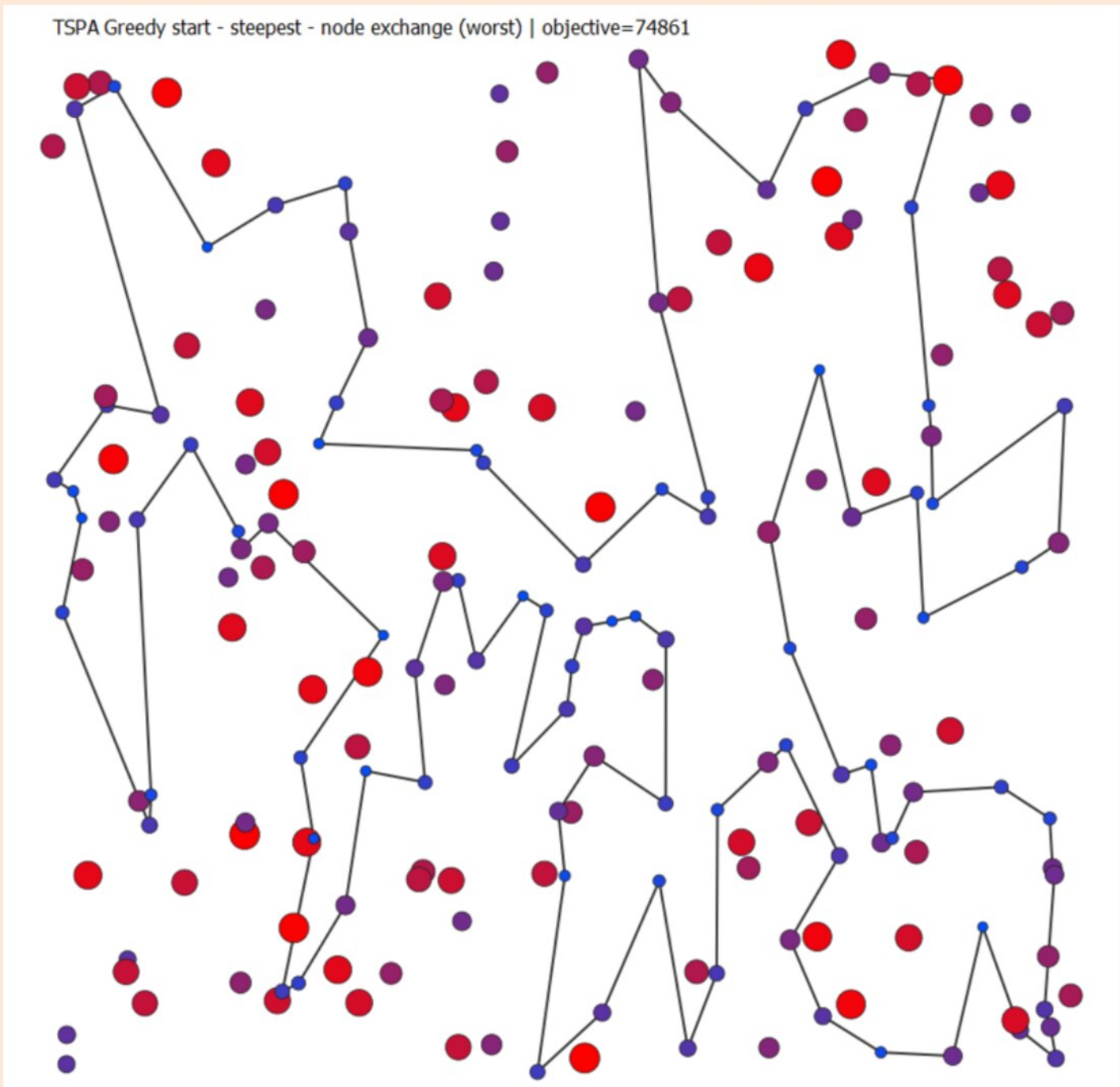
objective 80421

Worst solution nodes:

168 119 63 116 149 60 48
80 38 132 76 162 90 100
152 49 108 123 83 160 148
30 180 28 102 185 156 35
16 51 4 71 91 157 139 112
62 143 33 54 9 164 136
178 1 50 3 47 93 130 159
107 57 89 151 106 150 176
174 8 131 36 52 177 179
53 144 67 45 140 41 61 73
135 84 58 134 158 192 181
114 98 25 10 27 87 141
189 187 161 2 32 171 92
104 129 69 127 97 95 168

100 selected nodes; closed route

8. Large worst visualizations - TSPA



WORST SOLUTION

Best greedy / steepest / node exchange

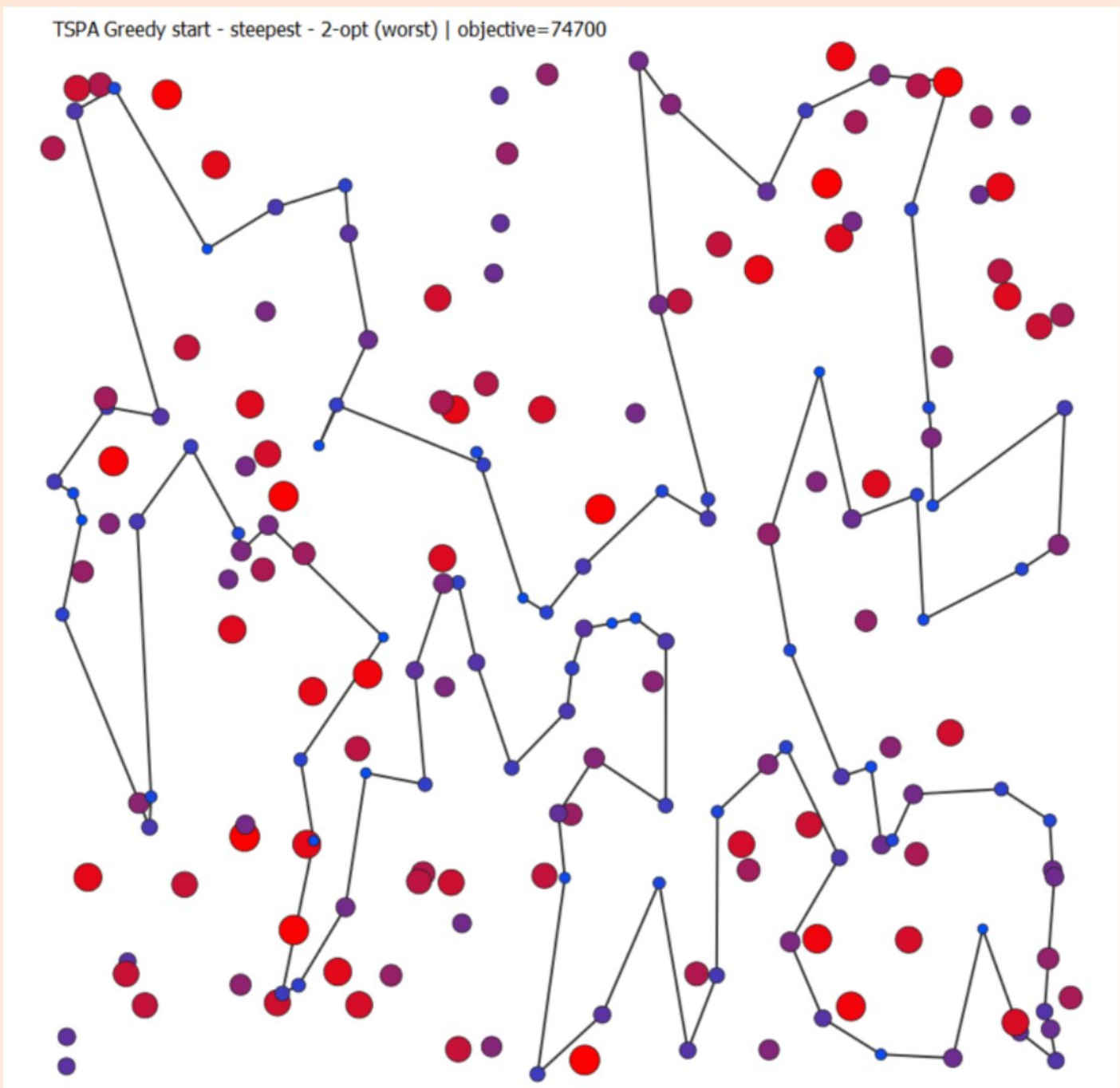
min=71955
max=74861
avg=72983.04

objective 74861

Worst solution nodes:

154 186 136 163 29 178 1
169 50 147 37 47 93 3 85
164 8 174 150 106 42 89
113 130 159 146 170 153
193 78 88 107 64 57 182
138 151 176 72 134 158
192 58 181 114 135 73 41
84 131 36 177 179 53 140
45 144 67 185 35 156 28
87 141 189 2 171 187 30
180 90 83 123 152 49 97
127 168 63 149 116 60 108
31 48 80 91 157 71 4 76
132 51 16 52 33 54 9 62
39 154

100 selected nodes; closed route



WORST SOLUTION

Best greedy / steepest / 2-opt

min=71955
max=74700
avg=73003.71

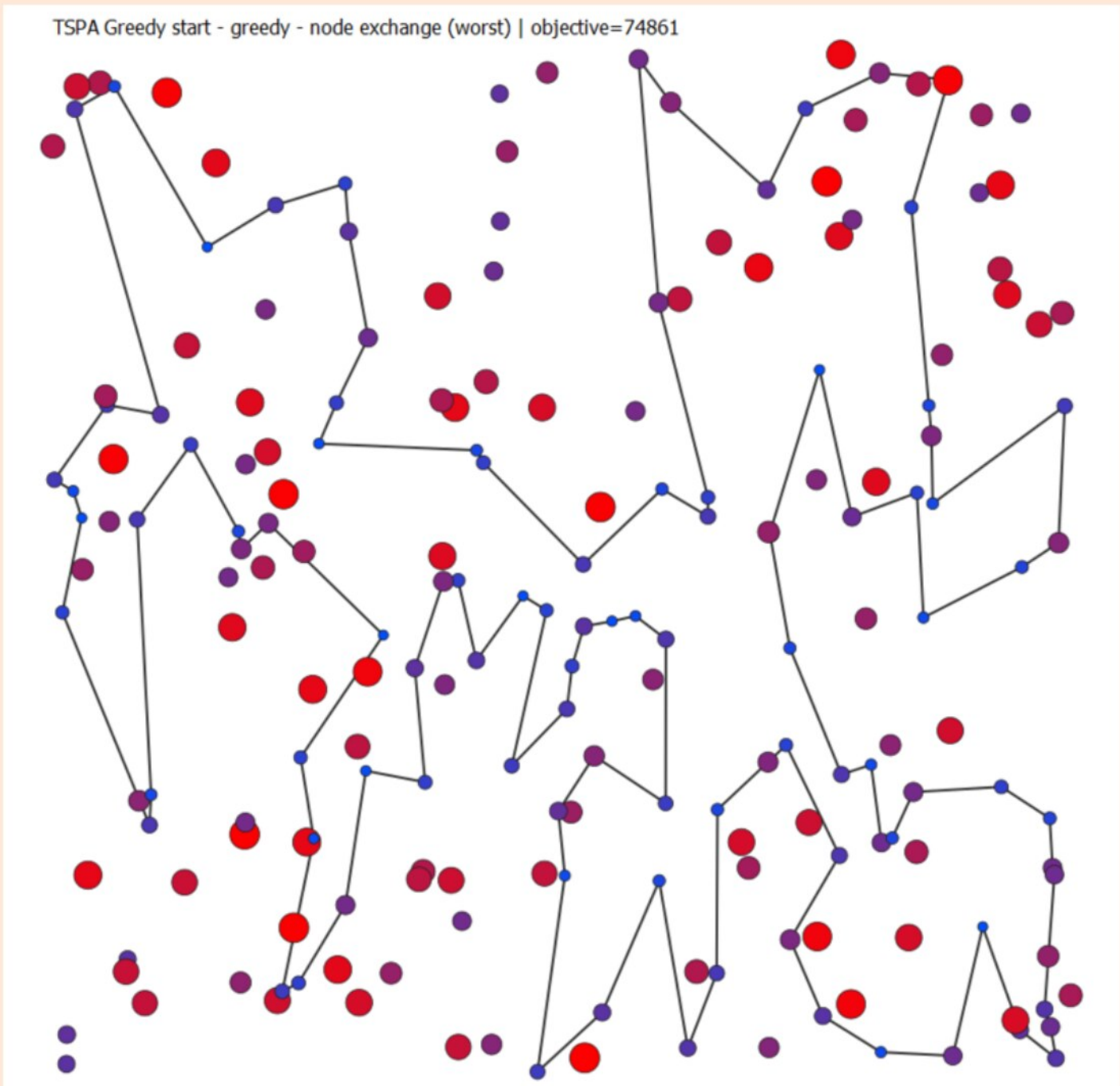
objective 74700

Worst solution nodes:

39 62 9 54 33 52 144 67
51 16 76 132 4 71 157 91
80 48 31 108 60 116 149
63 168 127 97 49 152 123
83 90 180 30 187 171 2
189 141 87 28 156 35 185
45 140 53 179 177 36 131
84 41 73 135 114 181 58
192 158 134 72 176 151
138 182 57 64 107 88 78
193 153 170 146 159 130
113 89 42 106 150 174 8
164 85 3 93 47 37 147 50
169 1 178 29 163 136 186
154 39

100 selected nodes; closed route

8. Large worst visualizations - TSPA



WORST SOLUTION

Best greedy / greedy / node exchange

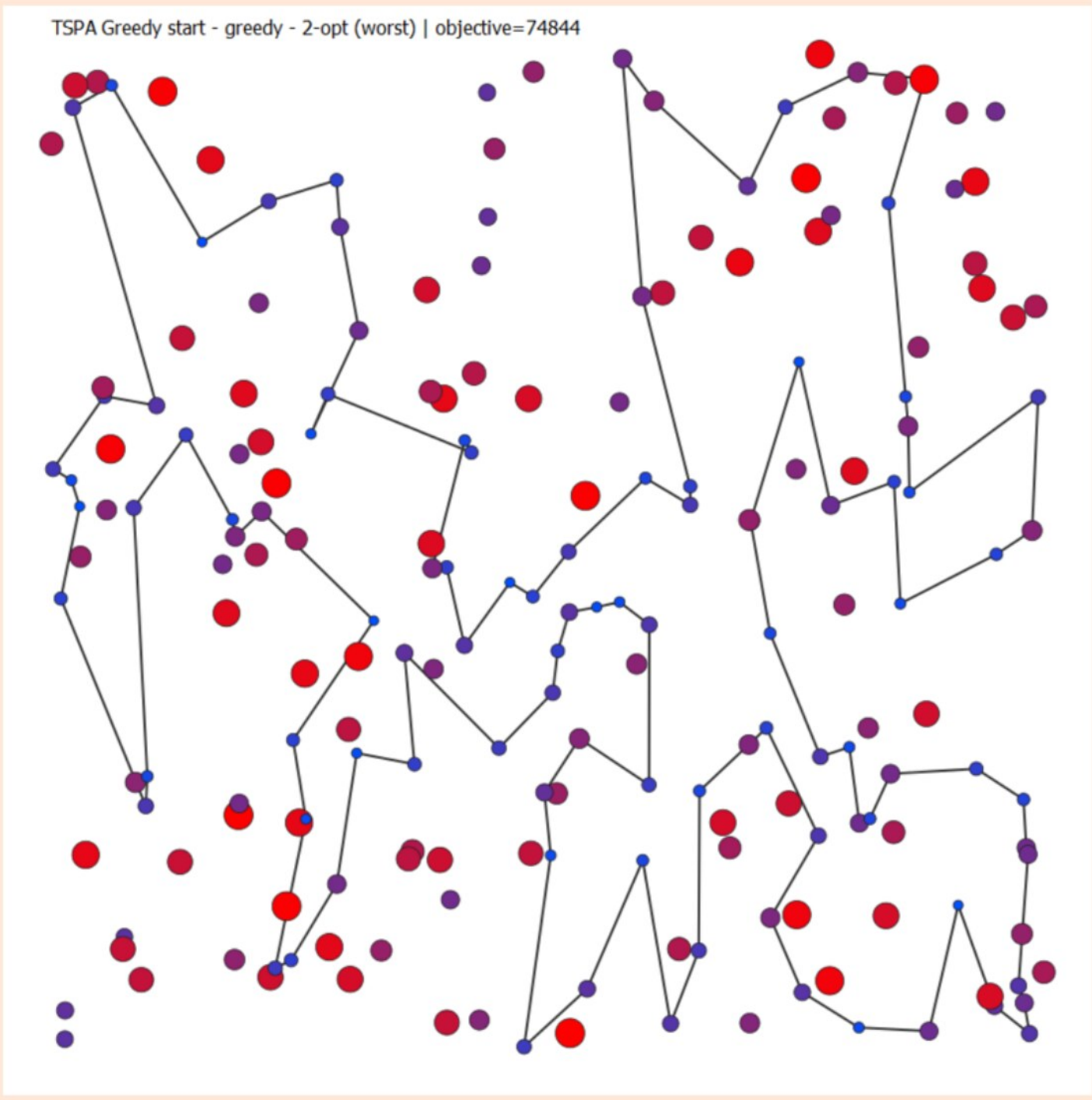
min=71955
max=74861
avg=72984.80

objective 74861

Worst solution nodes:

154 186 136 163 29 178 1
169 50 147 37 47 93 3 85
164 8 174 150 106 42 89
113 130 159 146 170 153
193 78 88 107 64 57 182
138 151 176 72 134 158
192 58 181 114 135 73 41
84 131 36 177 179 53 140
45 144 67 185 35 156 28
87 141 189 2 171 187 30
180 90 83 123 152 49 97
127 168 63 149 116 60 108
31 48 80 91 157 71 4 76
132 51 16 52 33 54 9 62
39 154

100 selected nodes; closed route



WORST SOLUTION

Best greedy / greedy / 2-opt

min=71955
max=74844
avg=73015.66

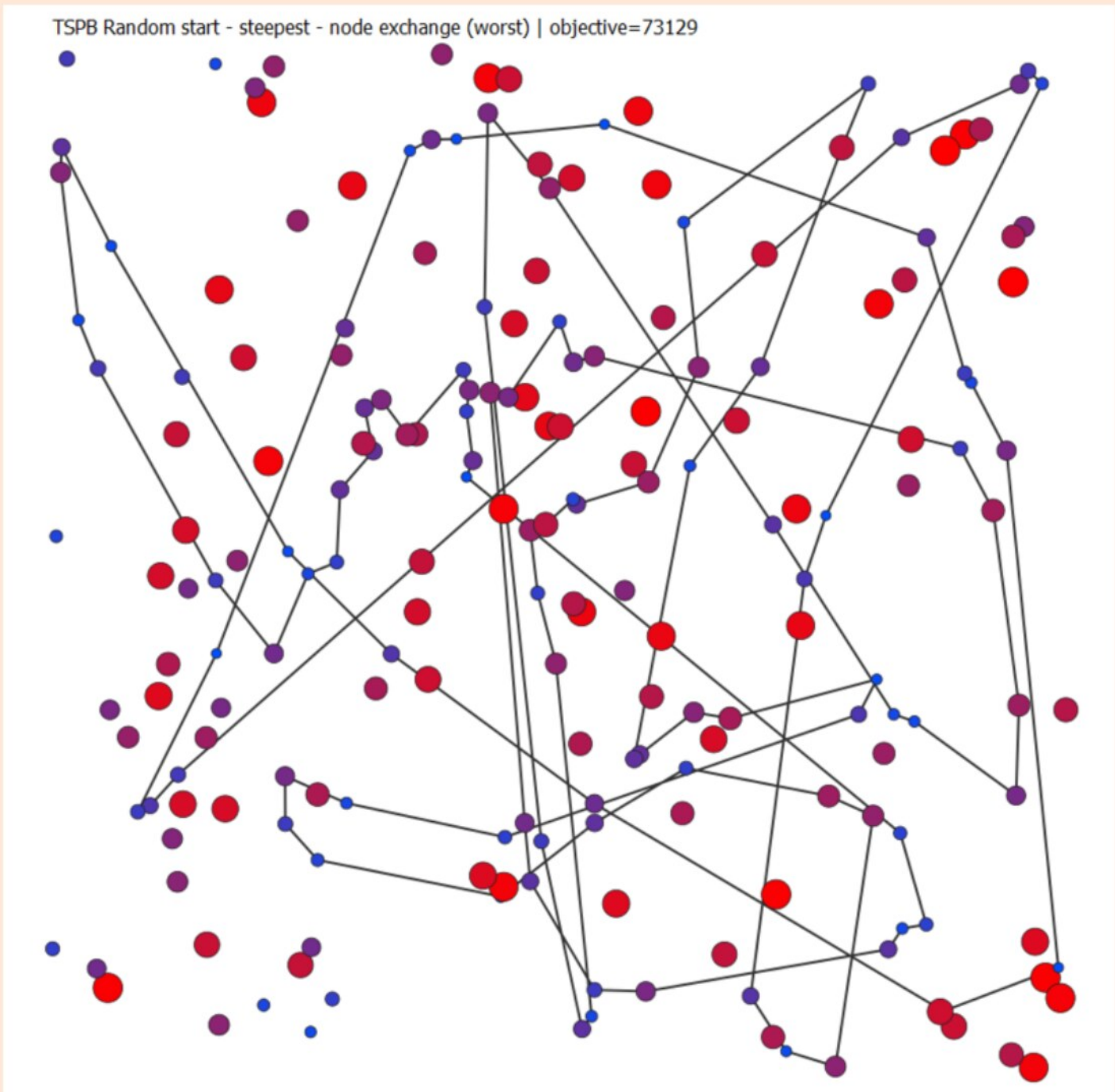
objective 74844

Worst solution nodes:

39 62 9 54 33 52 144 67
185 35 156 51 16 76 132 4
71 157 91 80 48 31 108 60
116 149 63 168 127 97 49
152 123 83 90 180 30 187
171 2 189 141 87 28 45
140 53 179 177 36 131 84
41 73 135 114 181 58 192
158 134 72 176 151 138
182 57 64 107 88 78 193
153 170 146 159 130 113
89 42 106 150 174 8 164
85 3 93 47 37 147 50 169
1 178 29 163 136 186 154
39

100 selected nodes; closed route

8. Large worst visualizations - TSPB



WORST SOLUTION

Random / steepest / node exchange

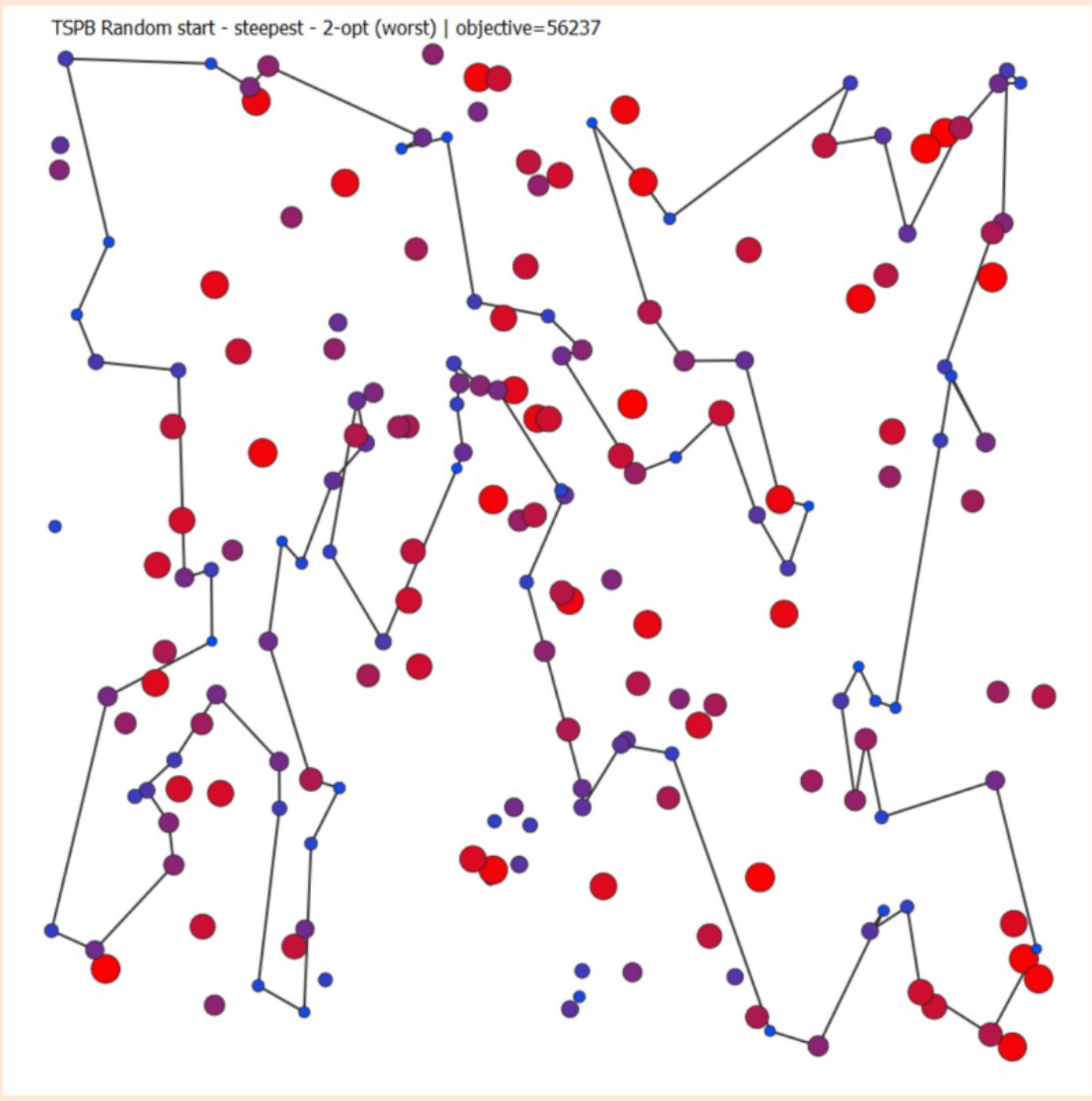
min=59057
max=73129
avg=65886.03

objective 73129

Worst solution nodes:

```
20 14 153 69 160 172 136
133 189 183 30 21 38 47
140 97 125 145 28 13 146
149 154 64 152 2 110 148
114 196 187 89 96 198 121
170 132 29 6 70 81 94 105
54 192 99 171 17 182 150
67 181 85 60 71 155 40 59
80 127 84 66 24 34 76 104
194 190 57 185 135 3 184
173 122 151 98 87 82 167
120 48 156 162 129 90 118
56 143 10 86 46 72 117 8
73 51 179 88 74 20
```

100 selected nodes; closed route



WORST SOLUTION

Random / steepest / 2-opt

min=50828
max=56237
avg=53457.16

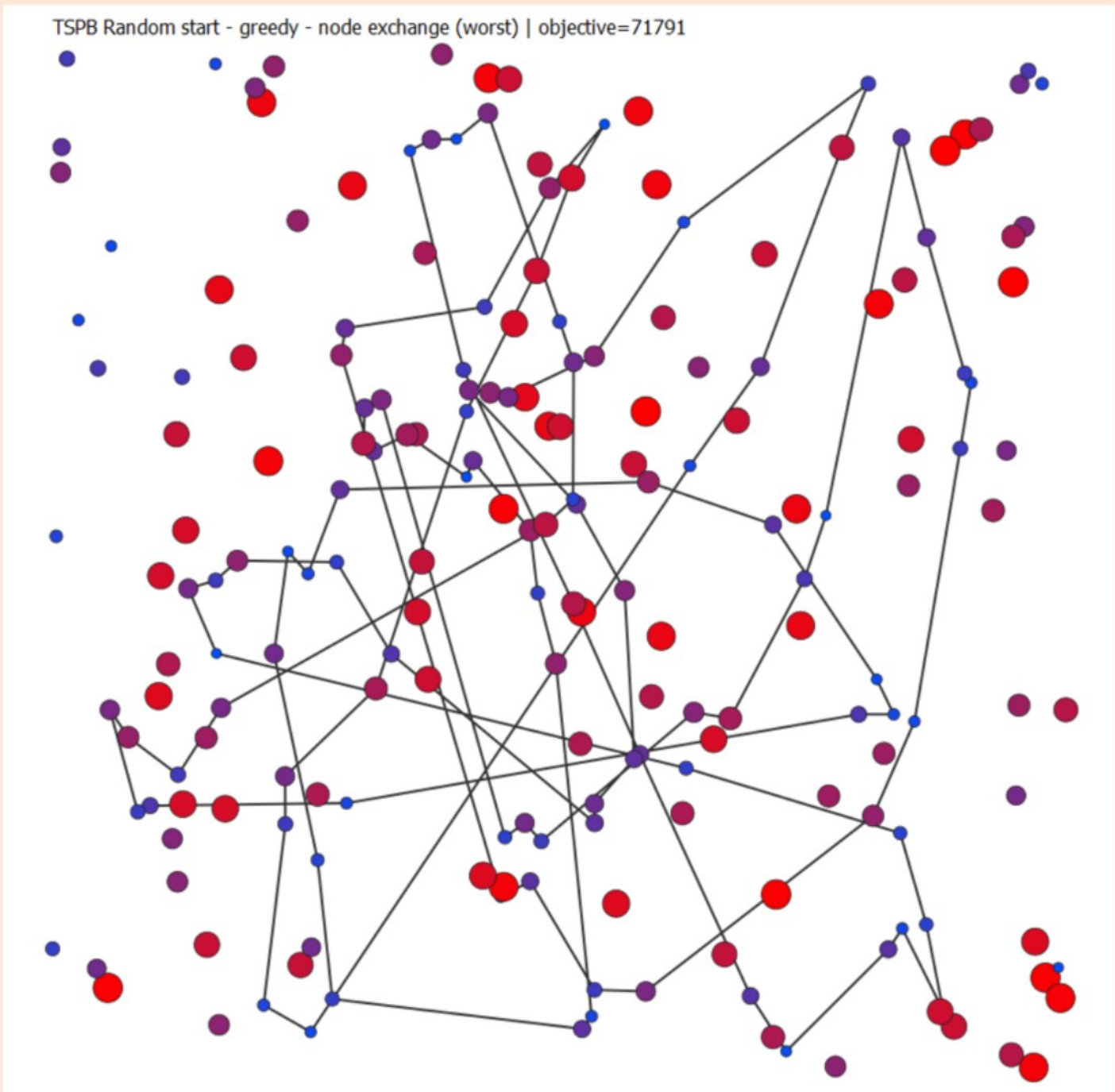
objective 56237

Worst solution nodes:

```
139 196 148 114 10 90 129
127 80 182 52 153 118 110
93 2 188 14 74 20 17 171
67 34 181 152 151 3 184
117 8 133 136 172 160 69
64 47 30 21 38 140 154 97
141 40 60 134 175 71 155
115 6 132 29 197 36 174 1
178 170 125 177 158 145
28 149 37 116 147 42 198
121 96 120 72 86 46 135
76 144 162 192 54 159 104
185 79 89 83 57 190 194
70 187 169 81 105 94 61
139
```

100 selected nodes; closed route

8. Large worst visualizations - TSPB



WORST SOLUTION

Random / greedy / node exchange

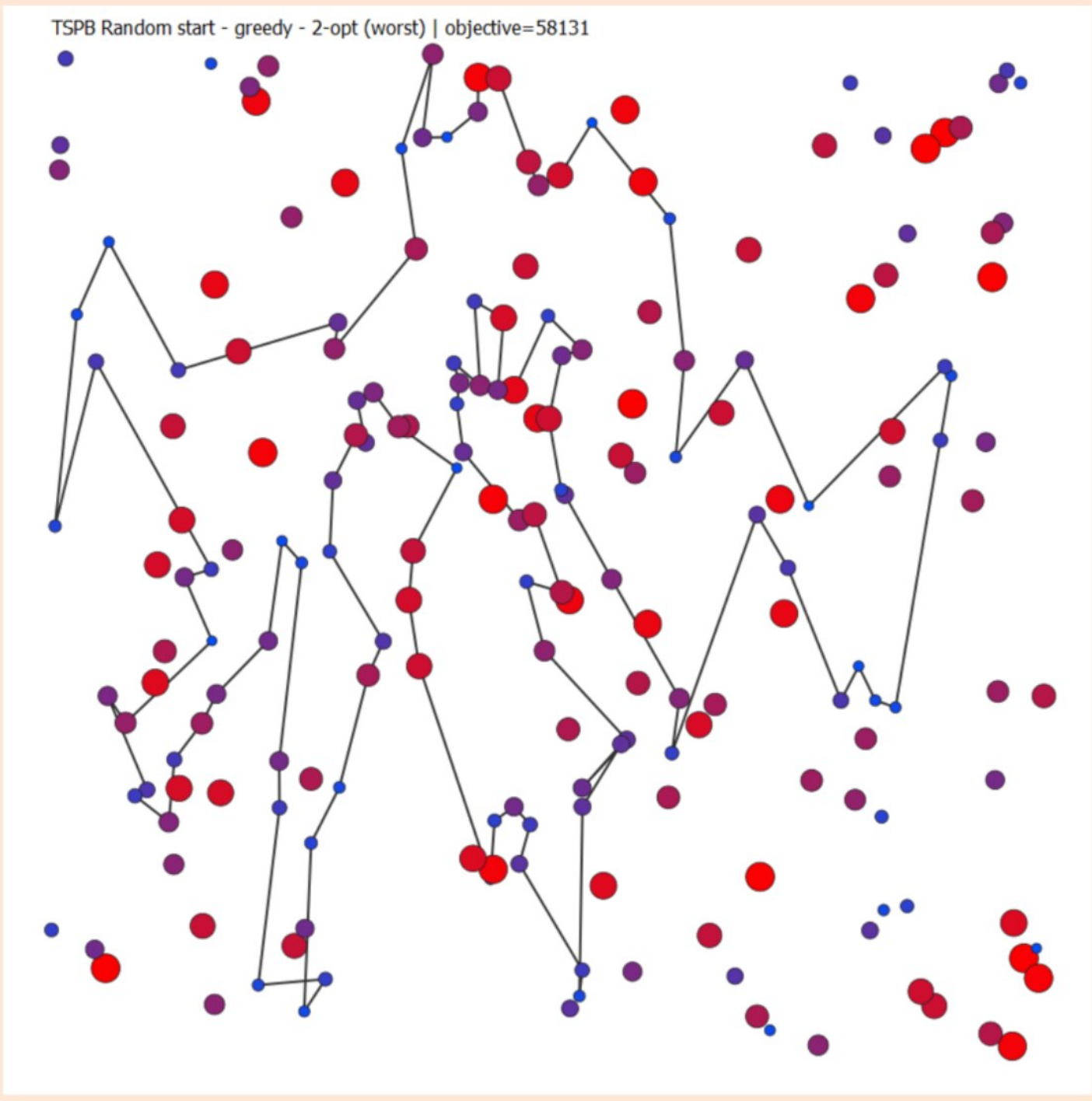
min=58625
max=71791
avg=63910.23

objective 71791

Worst solution nodes:

```
26 125 177 170 67 153 14
2 20 74 171 99 24 111 133
121 198 96 48 72 46 184
173 115 6 103 178 132 29
40 80 129 127 162 135 38
140 154 97 60 45 82 87 98
151 122 160 69 189 21 176
30 183 59 73 167 66 84
192 54 70 187 196 114 10
90 182 88 179 51 85 191
130 120 156 89 32 166 172
33 155 71 175 134 76 104
194 190 57 86 117 8 136 3
27 34 152 181 64 47 26
```

100 selected nodes; closed route



WORST SOLUTION

Random / greedy / 2-opt

min=51505
max=58131
avg=53797.47

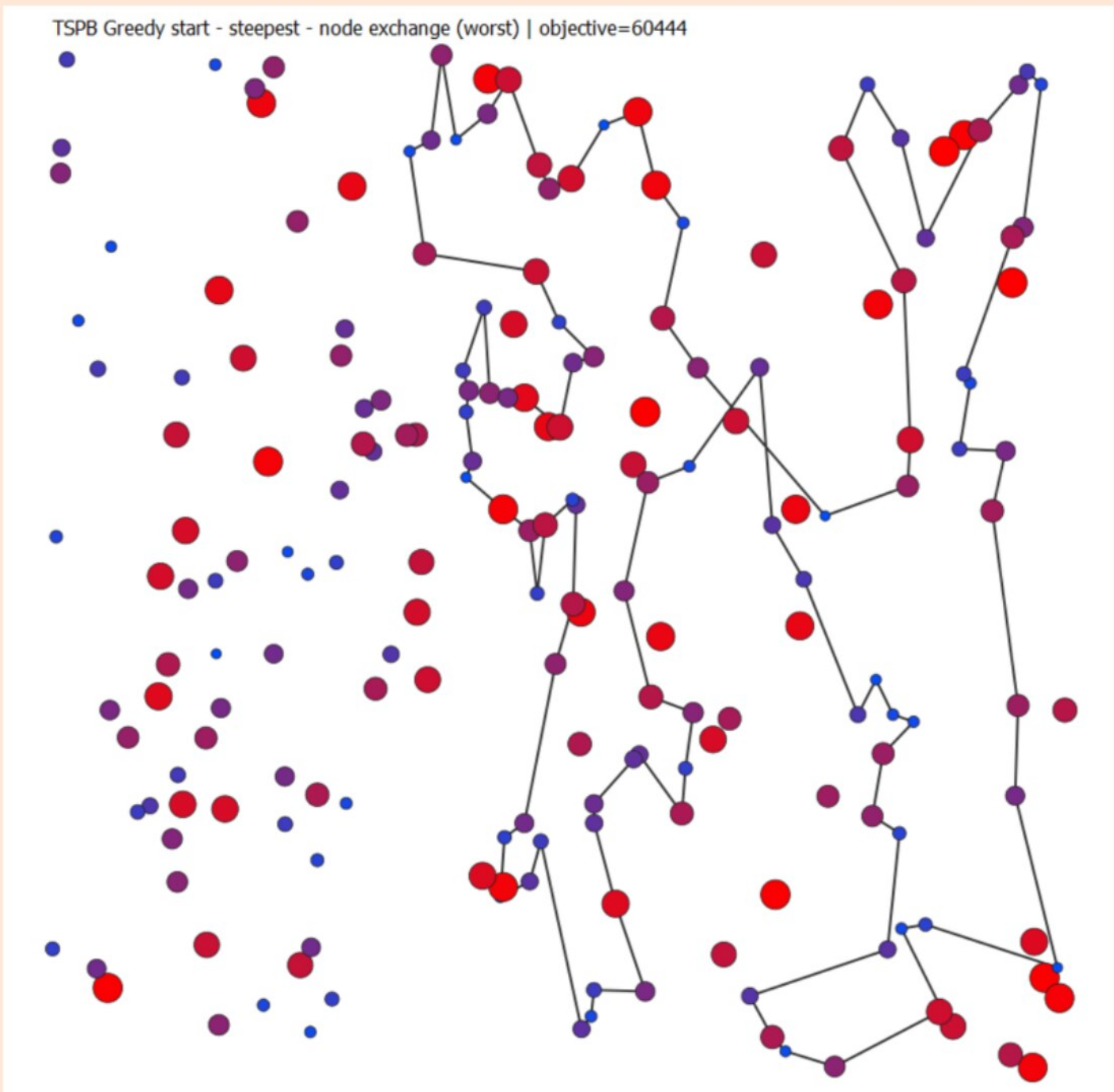
objective 58131

Worst solution nodes:

```
196 114 90 129 127 80 192
162 67 66 27 3 184 137 46
86 72 95 117 92 120 8 133
136 172 160 122 173 111
151 98 24 152 34 181 87
179 51 167 73 59 85 49 4
124 69 189 183 30 21 176
38 47 64 33 40 60 134 45
175 71 155 140 154 97 115
180 6 197 132 29 178 103
170 177 125 145 65 28 149
158 130 191 63 121 58 198
96 48 16 164 43 156 32 89
57 185 76 104 54 196
```

100 selected nodes; closed route

8. Large worst visualizations - TSPB



WORST SOLUTION

Best greedy / steepest / node exchange

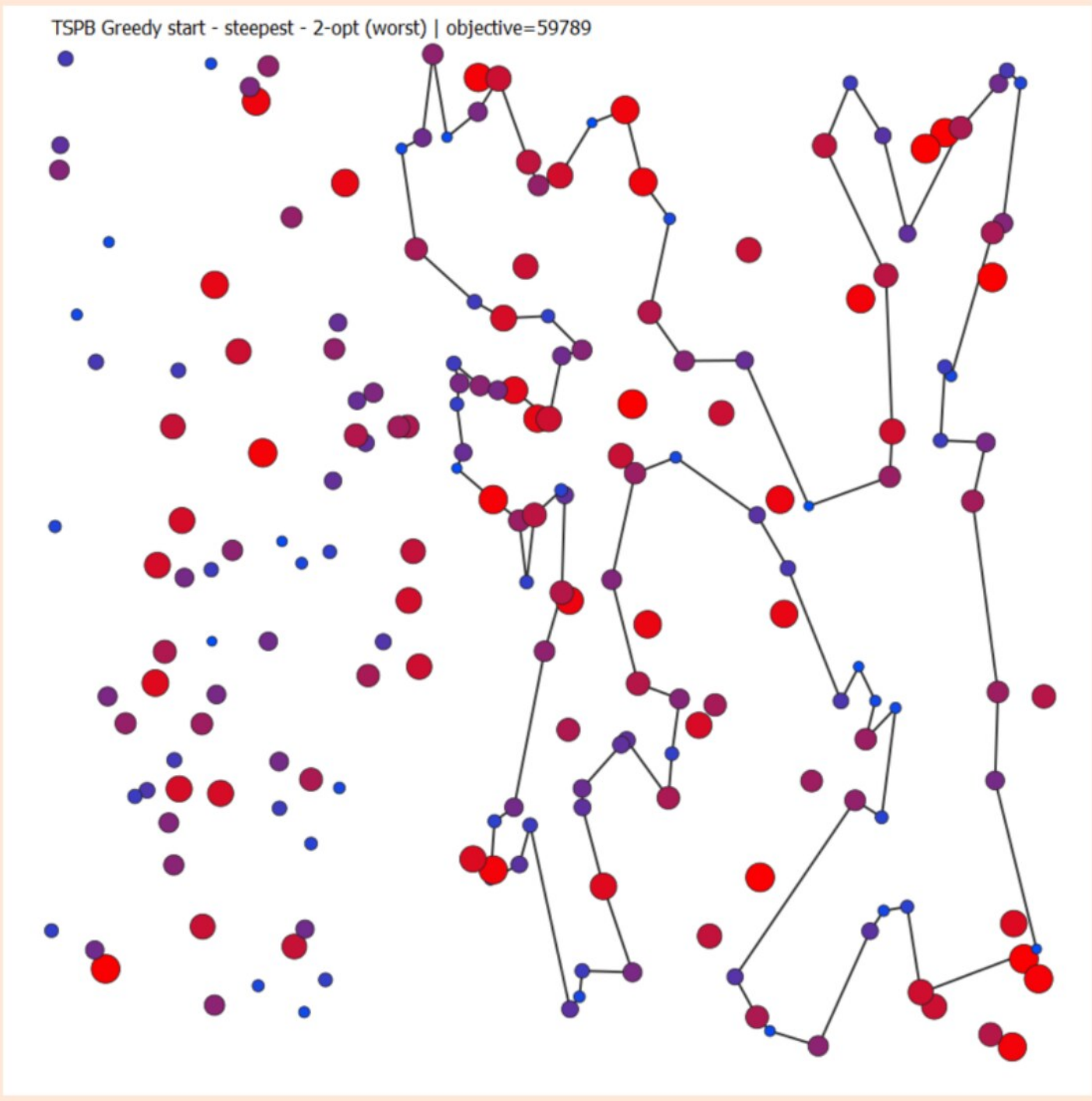
min=49785
max=60444
avg=53902.86

objective 60444

Worst solution nodes:

```
18 41 78 54 185 79 57 83
44 89 32 156 43 164 48 96
58 198 121 63 166 72 86
46 137 95 117 8 120 133
136 172 160 69 128 122
151 173 184 3 111 98 73
59 85 51 167 82 87 179 88
126 181 152 34 24 31 67
66 11 27 135 76 104 162
192 80 127 129 90 52 182
153 74 99 39 171 17 188 2
20 14 110 118 56 143 148
10 114 196 139 61 105 94
81 169 187 70 190 194 18
```

100 selected nodes; closed route



WORST SOLUTION

Best greedy / steepest / 2-opt

min=49676
max=59789
avg=53306.10

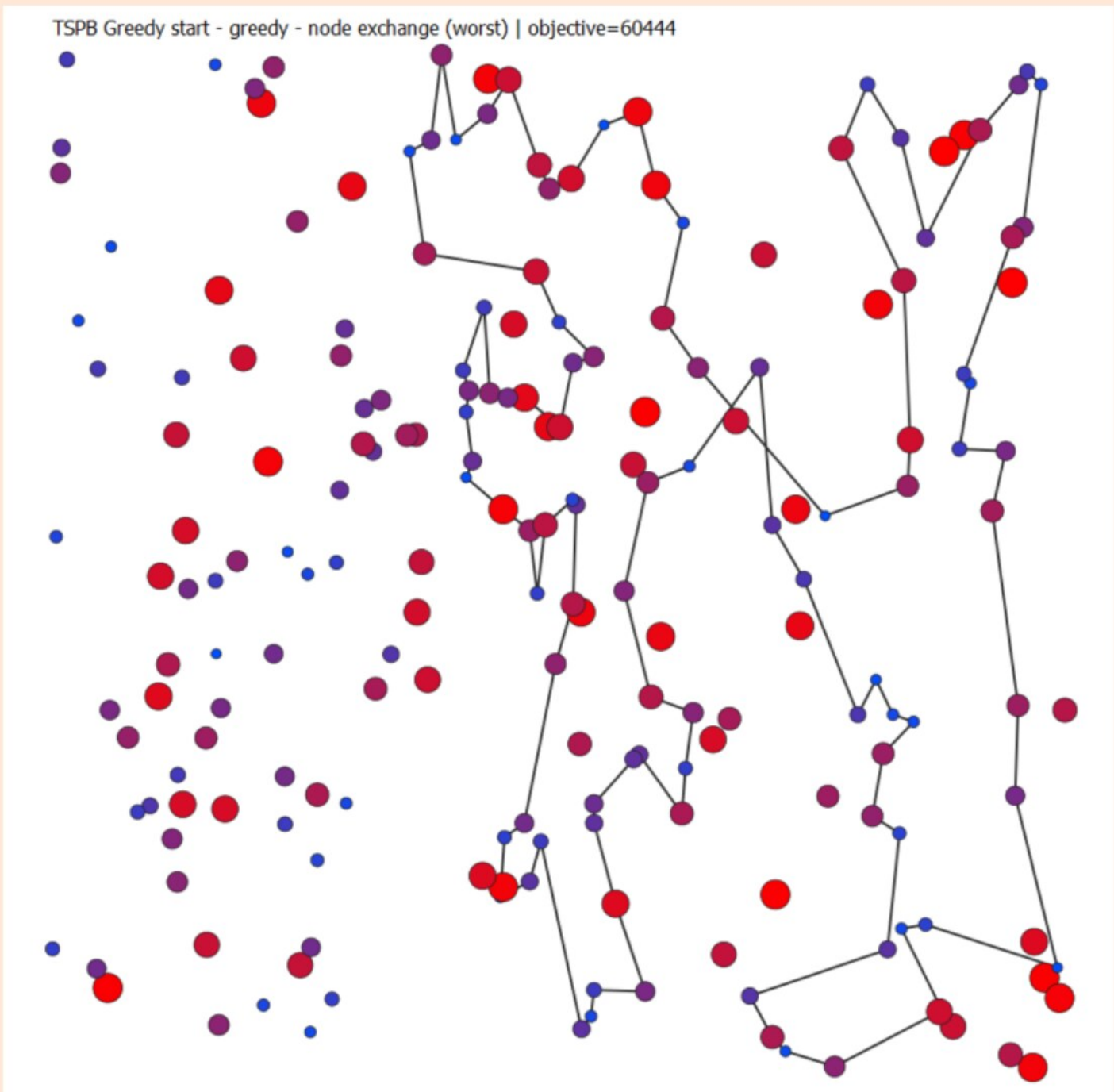
objective 59789

Worst solution nodes:

```
18 41 78 54 104 185 79 57
83 44 89 32 156 43 164 48
96 58 198 121 63 120 92
72 86 46 137 95 117 8 133
136 172 160 69 128 122
151 173 184 3 111 98 73
59 85 51 167 82 87 179 88
126 181 152 34 24 31 67
66 11 27 135 76 162 192
80 127 129 52 90 153 182
99 39 171 17 74 20 14 188
2 110 118 56 143 148 10
196 114 139 61 105 94 81
169 187 70 190 194 18
```

100 selected nodes; closed route

8. Large worst visualizations - TSPB



WORST SOLUTION

Best greedy / greedy / node exchange

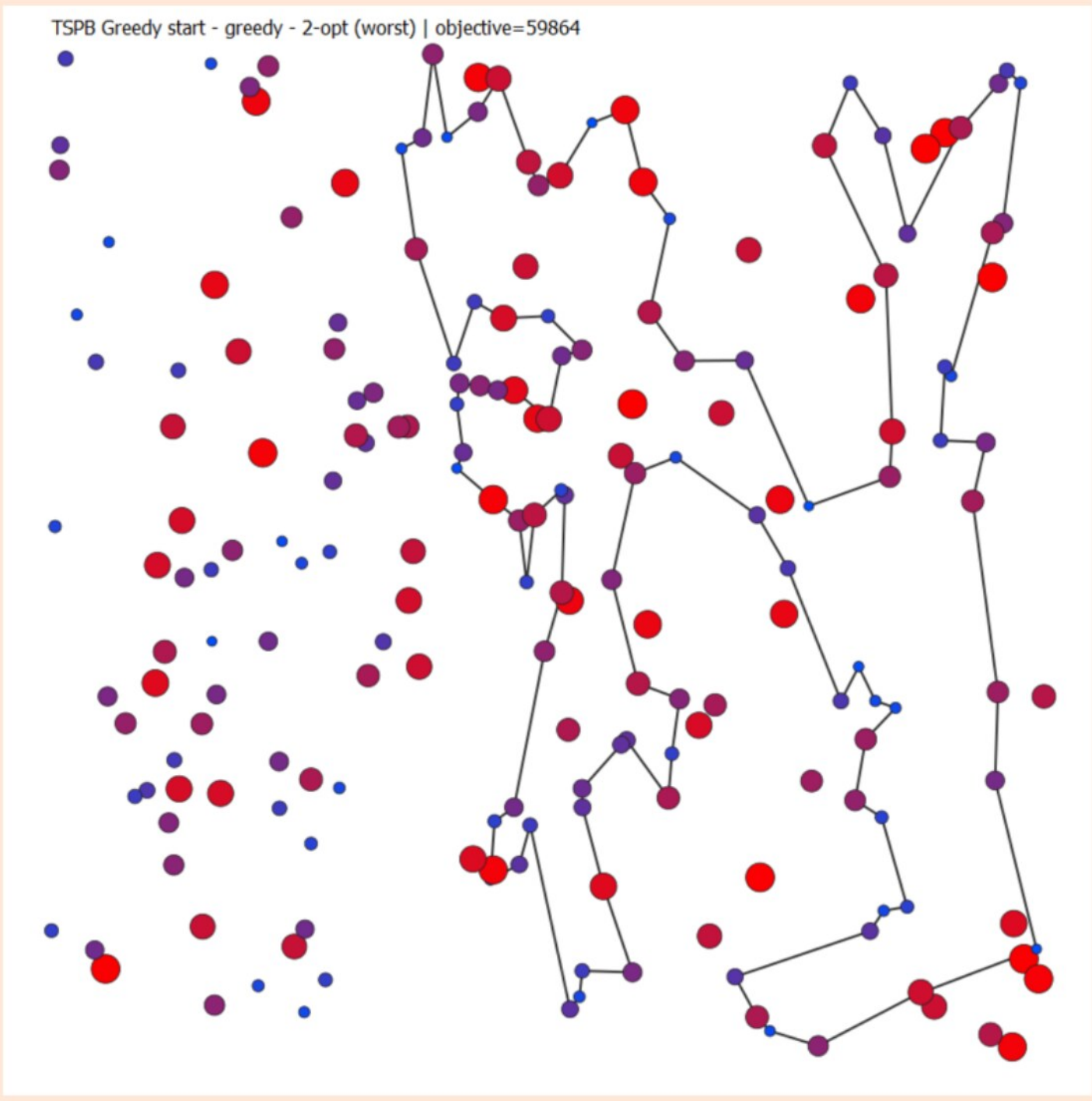
min=49849
max=60444
avg=53922.85

objective 60444

Worst solution nodes:

18 41 78 54 185 79 57 83
44 89 32 156 43 164 48 96
58 198 121 63 166 72 86
46 137 95 117 8 120 133
136 172 160 69 128 122
151 173 184 3 111 98 73
59 85 51 167 82 87 179 88
126 181 152 34 24 31 67
66 11 27 135 76 104 162
192 80 127 129 90 52 182
153 74 99 39 171 17 188 2
20 14 110 118 56 143 148
10 114 196 139 61 105 94
81 169 187 70 190 194 18

100 selected nodes; closed route



WORST SOLUTION

Best greedy / greedy / 2-opt

min=49710
max=59864
avg=53373.11

objective 59864

Worst solution nodes:

18 41 78 54 104 185 79 57
83 44 89 32 156 43 164 48
96 58 198 121 63 133 120
92 72 86 46 137 95 117 8
136 172 160 69 128 122
151 173 184 3 111 98 73
59 85 51 167 82 87 179 88
126 181 152 34 24 31 67
66 11 27 135 76 162 192
80 127 129 90 52 182 153
14 20 74 99 39 171 17 188
2 110 118 56 143 148 10
196 114 139 61 105 94 81
169 187 70 190 194 18

100 selected nodes; closed route